

Baboon Teeth Scott Scores

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Intro

Baboon teeth

Questions from Department Chair Krueger

Lastly, here are the questions: - Do wear scores increase with age? (Yes, duh, but we want to test it since we modified the method) - Do wear scores differ by sex? If so, how? (Is this by a specific quadrant? Is this by a side of tooth like buccal side or lingual side?) - Are there any patterns for rate of wear? That is, is there a standard wear through time (by month or year), or are some baboons wearing their teeth down faster than what we might expect?

Exploratory Data Analysis

Warning: NAs introduced by coercion

Summary statistics

```
library(tidyverse)
scott <- scott %>% mutate(diff = age_death - age_cast)

#by sex
scott %>% group_by(sex) %>%
  summarize(n = n(),
            mean_age_cast = mean(age_cast),
            mean_age_death = mean(age_death),
            mean_diff = mean(diff)) %>% as.data.frame()
```

	sex	n	mean_age_cast	mean_age_death	mean_diff
1	female	150	233.5333	262.9600	29.42667
2	male	51	196.0980	224.3725	28.27451

```
#overall
scott %>%
  summarize(n = n(),
            mean_age_cast = mean(age_cast),
            mean_age_death = mean(age_death),
            mean_diff = mean(diff))
```

	n	mean_age_cast	mean_age_death	mean_diff
1	201	224.0348	253.1692	29.13433

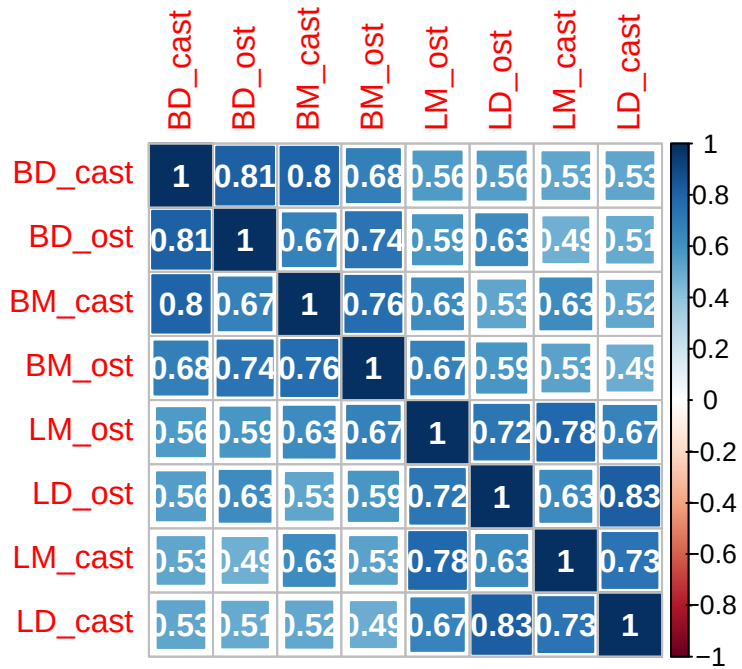
Variable summaries

```
summary(scott)
```

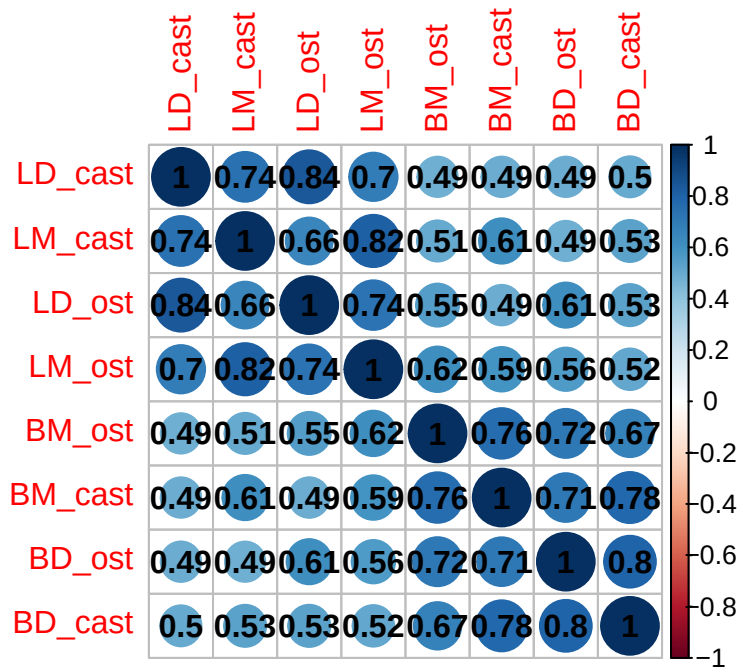
id		sex	age_cast		age_death				
Min.	: 3.0	Length:201	Min.	: 85	Min.	:102.0			
1st Qu.	:171.0	Class :character	1st Qu.	:185	1st Qu.	:215.0			
Median	:264.0	Mode :character	Median	:232	Median	:260.0			
Mean	:285.6		Mean	:224	Mean	:253.2			
3rd Qu.	:420.0		3rd Qu.	:256	3rd Qu.	:289.0			
Max.	:621.0		Max.	:401	Max.	:404.0			
sfbrid		leftright	BM_cast		BD_cast				
Length:	201	Length:201	Min.	:2.000	Min.	:1.000			
Class	:character	Class :character	1st Qu.	:5.000	1st Qu.	:5.000			
Mode	:character	Mode :character	Median	:6.000	Median	:5.000			
			Mean	:5.851	Mean	:5.308			
			3rd Qu.	:6.000	3rd Qu.	:6.000			
			Max.	:9.000	Max.	:9.000			
LM_cast		LD_cast	BM_ost		BD_ost		LM_ost		
Min.	:1.00	Min.	:1.000	Min.	:4.000	Min.	:1.00	Min.	:2.000
1st Qu.	:2.00	1st Qu.	:2.000	1st Qu.	:6.000	1st Qu.	:5.00	1st Qu.	:4.000
Median	:4.00	Median	:4.000	Median	:6.000	Median	:6.00	Median	:5.000
Mean	:4.09	Mean	:3.622	Mean	:6.502	Mean	:6.06	Mean	:5.348
3rd Qu.	:5.00	3rd Qu.	:5.000	3rd Qu.	:7.000	3rd Qu.	:7.00	3rd Qu.	:6.000
Max.	:9.00	Max.	:9.000	Max.	:9.000	Max.	:9.00	Max.	:9.000
LD_ost		diff							

Min.	:1.000	Min.	: 1.00
1st Qu.:	3.000	1st Qu.:	15.00
Median	:5.000	Median	:29.00
Mean	:4.687	Mean	:29.13
3rd Qu.:	6.000	3rd Qu.:	43.00
Max.	:9.000	Max.	:59.00

Correlations



Spearman correlations

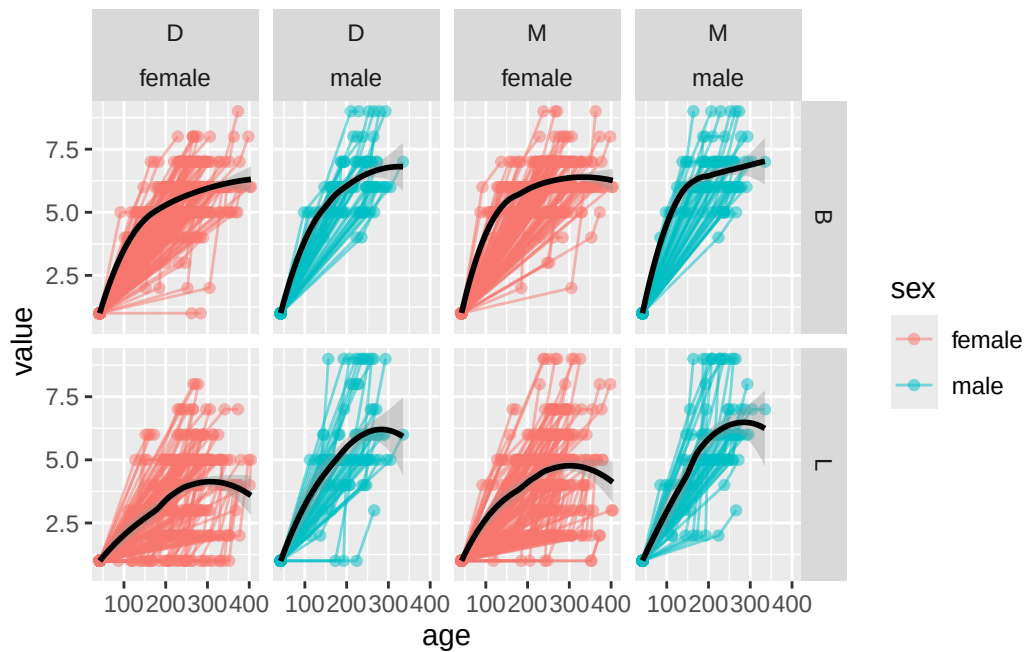


P-values for the Spearman correlation test

	BM_cast	BD_cast	LM_cast	LD_cast	BM_ost
BM_cast	NA	1.513686e-42	2.872028e-22	1.529116e-13	7.478033e-40
BD_cast	NA	NA	5.747780e-16	2.357984e-14	6.081874e-28
LM_cast	NA	NA	NA	9.839248e-37	1.634475e-14
LD_cast	NA	NA	NA	NA	2.202239e-13
BM_ost	NA	NA	NA	NA	NA
BD_ost	NA	NA	NA	NA	NA
LM_ost	NA	NA	NA	NA	NA
LD_ost	NA	NA	NA	NA	NA
	BD_ost	LM_ost	LD_ost		
BM_cast	3.032539e-32	1.418489e-20	1.157733e-13		
BD_cast	6.837721e-47	3.679490e-15	4.534340e-16		
LM_cast	2.431100e-13	1.678756e-49	2.500371e-26		
LD_cast	1.504922e-13	6.651144e-31	5.686140e-55		
BM_ost	2.609605e-33	1.056400e-22	1.370442e-17		
BD_ost	NA	2.924310e-18	5.951075e-22		
LM_ost	NA	NA	1.477813e-35		
LD_ost	NA	NA	NA		

The Bonferroni corrected cut off here is 0.0017857. All of these p-values are well below this cut-off.

Data Viz: Age vs wear by sex



Average and standard deviation of scott scores by quadrant

```
library(tidyverse)
scott_long %>% filter(castost != "eruption") %>% group_by(sex, castost, BLMD) %>% summarize(
```

`summarise()` has grouped output by 'sex', 'castost'. You can override using the `.groups` argument.

```
# A tibble: 16 x 5
# Groups:   sex, castost [4]
  sex  castost BLMD  mean  sd
<chr> <chr> <chr> <dbl> <dbl>
1 female cast BD 5.21 1.15
2 female cast BM 5.81 0.988
3 female cast LD 3.19 1.69
4 female cast LM 3.79 1.75
5 female ost BD 5.93 1.03
6 female ost BM 6.41 0.883
7 female ost LD 4.24 1.58
```

8	female	ost	LM	5.01	1.57
9	male	cast	BD	5.61	1.11
10	male	cast	BM	5.98	0.948
11	male	cast	LD	4.88	1.80
12	male	cast	LM	4.98	1.63
13	male	ost	BD	6.43	1.17
14	male	ost	BM	6.78	1.10
15	male	ost	LD	6	1.78
16	male	ost	LM	6.33	1.82

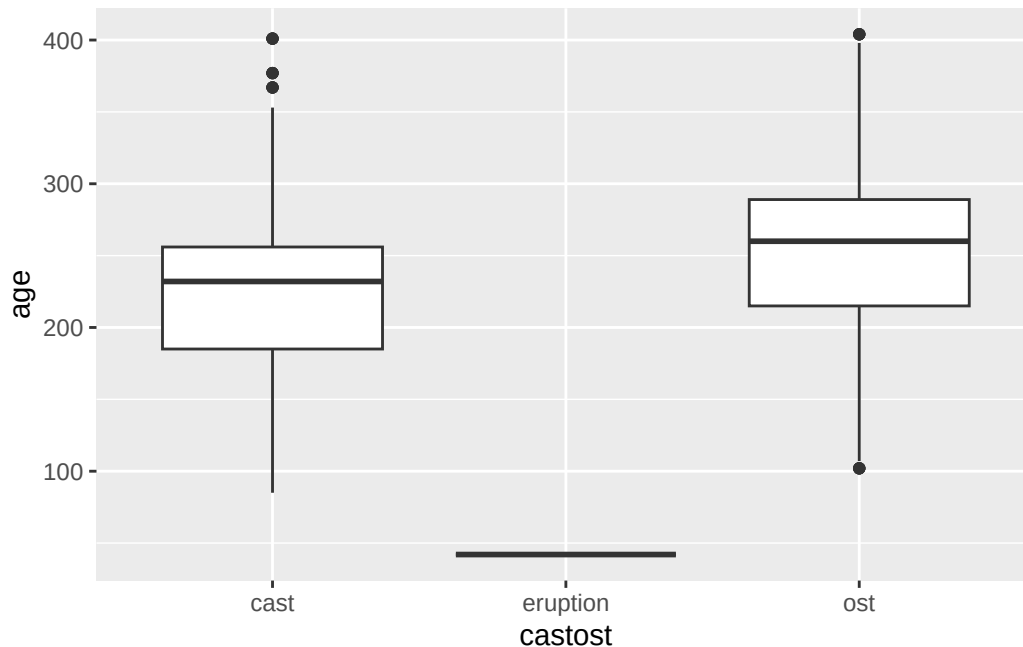
```
library(tidyverse)
scott_long %>% filter(castost != "eruption") %>% group_by(castost, BLMD) %>% summarize(mean =
```

`summarise()` has grouped output by 'castost'. You can override using the
`.groups` argument.

```
# A tibble: 8 x 4
# Groups:   castost [2]
  castost BLMD    mean    sd
  <chr>   <chr> <dbl> <dbl>
1 cast    BD      5.31  1.15
2 cast    BM      5.85  0.979
3 cast    LD      3.62  1.87
4 cast    LM      4.09  1.79
5 ost     BD      6.06  1.09
6 ost     BM      6.50  0.955
7 ost     LD      4.69  1.80
8 ost     LM      5.35  1.73
```

Age distributions by measurement time.

```
ggplot(aes(x = castost, y = age), data = scott_long %>% filter(age > 1)) + geom_boxplot() +
```



Do wear scores increase with age? (Yes, duh, but we want to test it since we modified the method)

The tests below test the following hypothesis: $H_0 : \mu_{cast} = \mu_{ost}$ vs $H_0 : \mu_{cast} < \mu_{ost}$ for each location of the tooth individually. The test used here is a two sample dependent t-test. In all cases, the null hypothesis is rejected with a p-value less than 2.2×10^{-16} . There is statistical evidence that the mean at ost is larger than the mean at cast.

```
t.test(scott$BM_cast, scott$BM_ost, paired = TRUE, alternative = "less")
```

Paired t-test

```
data:  scott$BM_cast and scott$BM_ost
t = -13.803, df = 200, p-value < 2.2e-16
alternative hypothesis: true mean difference is less than 0
95 percent confidence interval:
    -Inf -0.5737156
sample estimates:
mean difference
    -0.6517413
```

```
t.test(scott$BD_cast, scott$BD_ost, paired = TRUE, alternative = "less")
```

Paired t-test

```
data:  scott$BD_cast and scott$BD_ost
t = -15.408, df = 200, p-value < 2.2e-16
alternative hypothesis: true mean difference is less than 0
95 percent confidence interval:
    -Inf -0.6706737
sample estimates:
mean difference
    -0.7512438
```

```
t.test(scott$LM_cast, scott$LM_ost, paired = TRUE, alternative = "less")
```

Paired t-test

```
data:  scott$LM_cast and scott$LM_ost
t = -15.343, df = 200, p-value < 2.2e-16
alternative hypothesis: true mean difference is less than 0
95 percent confidence interval:
    -Inf -1.12314
sample estimates:
mean difference
    -1.258706
```

```
t.test(scott$LD_cast, scott$LD_ost, paired = TRUE, alternative = "less")
```

Paired t-test

```
data:  scott$LD_cast and scott$LD_ost
t = -14.071, df = 200, p-value < 2.2e-16
alternative hypothesis: true mean difference is less than 0
95 percent confidence interval:
    -Inf -0.9396378
sample estimates:
mean difference
    -1.064677
```


No matter what correction you use here, these are going to be highly significant.

Modeling effect of age

Here are pearson correlation values for you. But the relationship clearly isn't linear so these probably aren't what you want. I've also computed Spearman correlation for you, which tests for any kind of monotonic relationship. This is much closer to what you are looking for than Pearson's coefficient.

Pearson correlation tests

I would argue there are 8 tests here in this family so the Bonferroni corrected cut off would be 0.00625.

```
#Pearson correlation
cor.test(scott$BM_cast,scott$age_cast)
```

Pearson's product-moment correlation

```
data:  scott$BM_cast and scott$age_cast
t = 3.1426, df = 199, p-value = 0.00193
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.08150242 0.34544369
sample estimates:
      cor
0.2174445
```

```
cor.test(scott$BD_cast,scott$age_cast)
```

Pearson's product-moment correlation

```
data:  scott$BD_cast and scott$age_cast
t = 4.6134, df = 199, p-value = 7.086e-06
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.1801893 0.4306997
sample estimates:
```

```
cor
0.3108327
```

```
cor.test(scott$LM_cast,scott$age_cast)
```

Pearson's product-moment correlation

```
data:  scott$LM_cast and scott$age_cast
t = 1.5346, df = 199, p-value = 0.1265
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.03070694  0.24290650
sample estimates:
      cor
0.1081474
```

```
cor.test(scott$LD_cast,scott$age_cast)
```

Pearson's product-moment correlation

```
data:  scott$LD_cast and scott$age_cast
t = 1.258, df = 199, p-value = 0.2098
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.05018341  0.22446354
sample estimates:
      cor
0.08882823
```

```
cor.test(scott$BM_ost,scott$age_death)
```

Pearson's product-moment correlation

```
data:  scott$BM_ost and scott$age_death
t = 2.2123, df = 199, p-value = 0.02808
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
```

```
0.01690182 0.28717130
sample estimates:
      cor
0.1549341
```

```
cor.test(scott$BD_ost,scott$age_death)
```

Pearson's product-moment correlation

```
data:  scott$BD_ost and scott$age_death
t = 4.0452, df = 199, p-value = 7.477e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.1426931 0.3988249
sample estimates:
      cor
0.2756444
```

```
cor.test(scott$LM_ost,scott$age_death)
```

Pearson's product-moment correlation

```
data:  scott$LM_ost and scott$age_death
t = 0.99168, df = 199, p-value = 0.3226
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
-0.06893834 0.20651590
sample estimates:
      cor
0.07012542
```

```
cor.test(scott$LD_ost,scott$age_death)
```

Pearson's product-moment correlation

```
data:  scott$LD_ost and scott$age_death
t = 1.1993, df = 199, p-value = 0.2318
```

```
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.0543204  0.2205206
sample estimates:
      cor
0.08471114
```

Spearman correlation tests

```
#Spearman correlation
cor.test(scott$BM_cast,scott$age_cast, method = "spearman")
```

Spearman's rank correlation rho

```
data:  scott$BM_cast and scott$age_cast
S = 970966, p-value = 4.817e-05
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
0.282573
```

```
cor.test(scott$BD_cast,scott$age_cast, method = "spearman")
```

Spearman's rank correlation rho

```
data:  scott$BD_cast and scott$age_cast
S = 875455, p-value = 2.717e-07
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
0.3531437
```

```
cor.test(scott$LM_cast,scott$age_cast, method = "spearman")
```

Spearman's rank correlation rho

```
data:  scott$LM_cast and scott$age_cast
S = 1164898, p-value = 0.04862
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
0.1392802
```

```
cor.test(scott$LD_cast,scott$age_cast, method = "spearman")
```

Spearman's rank correlation rho

```
data:  scott$LD_cast and scott$age_cast
S = 1211368, p-value = 0.1382
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
0.1049448
```

```
#Corelation
cor.test(scott$BM_ost,scott$age_death, method = "spearman")
```

Spearman's rank correlation rho

```
data:  scott$BM_ost and scott$age_death
S = 1146997, p-value = 0.03067
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
0.1525068
```

```
cor.test(scott$BD_ost,scott$age_death, method = "spearman")
```

Spearman's rank correlation rho

```
data:  scott$BD_ost and scott$age_death
S = 946152, p-value = 1.421e-05
alternative hypothesis: true rho is not equal to 0
```

sample estimates:

rho

0.3009077

```
cor.test(scott$LM_ost,scott$age_death, method = "spearman")
```

Spearman's rank correlation rho

data: scott\$LM_ost and scott\$age_death

S = 1245286, p-value = 0.2596

alternative hypothesis: true rho is not equal to 0

sample estimates:

rho

0.07988334

```
cor.test(scott$LD_ost,scott$age_death, method = "spearman")
```

Spearman's rank correlation rho

data: scott\$LD_ost and scott\$age_death

S = 1213542, p-value = 0.1443

alternative hypothesis: true rho is not equal to 0

sample estimates:

rho

0.1033379

P-values for relationship between wear and age.

I would argue there are 8 tests here in this family so the Bonferroni corrected cut off would be 0.00625. Using the cut-off, the tests that are significant are for BM cast and age, BD cast and age, and BD ost and age.

```
out %>% pivot_wider(names_from = "BDLM", values_from = "pvalue")
```

A tibble: 2 x 5

	castost	BM	BD	LM	LD
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	cast	0.0000482	0.000000272	0.0486	0.138
2	ost	0.0307	0.0000142	0.260	0.144

Do wear scores differ by sex? If so, how? (Is this by a specific quadrant? Is this by a side of tooth like buccal side or lingual side?)

If we ignore age entirely, there are significant differences in the mean vector of wear for males vs females at both cast and ost. If we then do individual tests for wear differences by region at cast and ost separately, there are significant differences in LM and LD at cast and LM at ost. (I used a 0.05 level of significant and a Bonferroni correction with 8 tests to get a cut off for significance of 0.00625. This is, as always, conservative.)

```
#Look for differences in male vs female with a hotelling t test
library(Hotelling)
```

Loading required package: corpcor

Attaching package: 'Hotelling'

The following object is masked from 'package:dplyr':

summarise

```
results_cast <- hotelling.test(~sex, data = scott %>% select(sex, BM_cast, BD_cast, LM_cast,
results_cast
```

Test stat: 42.938
Numerator df: 4
Denominator df: 86.9802290794176
P-value: 4.999e-07

```
results_ost <- hotelling.test(~sex, data = scott %>% select(sex, BM_ost, BD_ost, LM_ost, LD_
results_ost
```

Test stat: 46.791
Numerator df: 4
Denominator df: 77.4160522734112
P-value: 2.173e-07

```
#Now do individual t-tests on each component  
t.test(BM_cast~sex,data = scott)
```

Welch Two Sample t-test

```
data: BM_cast by sex  
t = -1.118, df = 89.591, p-value = 0.2665  
alternative hypothesis: true difference in means between group female and group male is not equal to 0  
95 percent confidence interval:  
 -0.4824452  0.1349942  
sample estimates:  
mean in group female    mean in group male  
      5.806667           5.980392
```

```
t.test(BD_cast~sex,data = scott)
```

Welch Two Sample t-test

```
data: BD_cast by sex  
t = -2.2027, df = 88.72, p-value = 0.03021  
alternative hypothesis: true difference in means between group female and group male is not equal to 0  
95 percent confidence interval:  
 -0.76308075 -0.03927219  
sample estimates:  
mean in group female    mean in group male  
      5.206667           5.607843
```

```
t.test(LD_cast~sex,data = scott)
```

Welch Two Sample t-test

```
data: LD_cast by sex  
t = -5.8849, df = 82.279, p-value = 8.303e-08  
alternative hypothesis: true difference in means between group female and group male is not equal to 0  
95 percent confidence interval:  
 -2.259947 -1.118092  
sample estimates:
```


mean in group female	mean in group male
3.193333	4.882353

```
t.test(LM_cast~sex,data = scott)
```

Welch Two Sample t-test

```
data: LM_cast by sex
t = -4.4302, df = 92.159, p-value = 2.589e-05
alternative hypothesis: true difference in means between group female and group male is not equal to 0
95 percent confidence interval:
 -1.728868 -0.658583
sample estimates:
mean in group female mean in group male
      3.786667      4.980392
```

```
#Now do individual t-tests on each component for ost
t.test(BM_ost~sex,data = scott)
```

Welch Two Sample t-test

```
data: BM_ost by sex
t = -2.2186, df = 73.084, p-value = 0.02962
alternative hypothesis: true difference in means between group female and group male is not equal to 0
95 percent confidence interval:
 -0.71688510 -0.03840902
sample estimates:
mean in group female mean in group male
      6.406667      6.784314
```

```
t.test(BD_ost~sex,data = scott)
```

Welch Two Sample t-test

```
data: BD_ost by sex
t = -2.7012, df = 78.207, p-value = 0.008468
alternative hypothesis: true difference in means between group female and group male is not equal to 0
```

```

95 percent confidence interval:
 -0.8650942 -0.1309843
sample estimates:
mean in group female    mean in group male
      5.933333           6.431373

```

```
t.test(LD_ost~sex,data = scott)
```

Welch Two Sample t-test

```

data: LD_ost by sex
t = -6.2787, df = 78.517, p-value = 1.755e-08
alternative hypothesis: true difference in means between group female and group male is not 0
95 percent confidence interval:
 -2.318006 -1.201994
sample estimates:
mean in group female    mean in group male
      4.24              6.00

```

```
t.test(LM_ost~sex,data = scott)
```

Welch Two Sample t-test

```

data: LM_ost by sex
t = -4.6294, df = 76.952, p-value = 1.462e-05
alternative hypothesis: true difference in means between group female and group male is not 0
95 percent confidence interval:
 -1.8877776 -0.7522224
sample estimates:
mean in group female    mean in group male
      5.013333           6.333333

```

Modeling

Scott Scores

But we should really be controlling for sex and age and id all at once and I think we should be looking at age at eruption with wear of 0.

Need to use REML = FALSE to compare model. Reason why: -<https://stats.stackexchange.com/questions/272654-to-decide-whether-to-set-reml-to-true-or-false/272654#272654> - <https://stackoverflow.com/questions/54980399/should-i-specify-reml-false-in-lmer>

Scott Score overall

```
library(lme4)
```

Loading required package: Matrix

Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

expand, pack, unpack

```
scott_long_sum <- scott_long %>% group_by(id, castost, age, sex) %>% summarize(value = sum(v
```

`summarise()` has grouped output by 'id', 'castost', 'age'. You can override using the `.groups` argument.

```
scott_long_sum %>% group_by(castost, sex) %>% summarize(mean = mean(value), std = sd(value))
```

`summarise()` has grouped output by 'castost'. You can override using the `.groups` argument.

```
# A tibble: 6 x 4
# Groups:   castost [3]
  castost sex    mean  std
  <chr>   <chr> <dbl> <dbl>
1 cast   female  18.0  4.70
2 cast   male    21.5  4.70
3 eruption female    4    0
4 eruption male     4    0
5 ost     female  21.6  4.35
6 ost     male    25.5  4.98
```

```
scott_long_sum %>% group_by(castost) %>% summarize(mean = mean(value), std = sd(value))
```

```
# A tibble: 3 x 3
  castost mean std
  <chr>   <dbl> <dbl>
1 cast    18.9  4.92
2 eruption  4    0
3 ost     22.6  4.82
```

```
mod_all_full <- lmer(value ~ (age+I(age^2) + I(age^3))*sex + (1|id), data = scott_long_sum, REML = FALSE)
```

Warning: Some predictor variables are on very different scales: consider rescaling

```
mod_all_red <- lmer(value ~ (age+I(age^2)+ I(age^3)) + (1|id), data = scott_long_sum, REML = FALSE)
```

Warning: Some predictor variables are on very different scales: consider rescaling

```
#ICC Overall
```

```
(VarCorr(mod_all_full)$id[1])/(VarCorr(mod_all_full)$id[1]+attr(VarCorr(mod_all_full),"sc"))
```

```
[1] 0.3322012
```

```
#test stat
```

```
-2*(logLik(mod_all_red) - logLik(mod_all_full))
```

```
'log Lik.' 103.8104 (df=6)
```

```
#p-value
```

```
1 - pchisq(-2*(logLik(mod_all_red) - logLik(mod_all_full)),4)
```

```
'log Lik.' 0 (df=6)
```

```
summary(mod_all_full)
```

Linear mixed model fit by maximum likelihood ['lmerMod']
 Formula: value ~ (age + I(age^2) + I(age^3)) * sex + (1 | id)
 Data: scott_long_sum

AIC	BIC	logLik	deviance	df.resid
3250.6	3294.7	-1615.3	3230.6	593

Scaled residuals:

Min	1Q	Median	3Q	Max
-3.2130	-0.5006	-0.0202	0.5130	3.0428

Random effects:

Groups	Name	Variance	Std.Dev.
id	(Intercept)	4.559	2.135
	Residual	9.164	3.027

Number of obs: 603, groups: id, 201

Fixed effects:

	Estimate	Std. Error	t value
(Intercept)	-3.513e+00	9.397e-01	-3.738
age	2.020e-01	2.559e-02	7.894
I(age^2)	-5.553e-04	1.500e-04	-3.703
I(age^3)	5.223e-07	2.448e-07	2.134
sexmale	-3.662e+00	2.178e+00	-1.681
age:sexmale	1.043e-01	6.333e-02	1.648
I(age^2):sexmale	-4.483e-04	4.261e-04	-1.052
I(age^3):sexmale	6.910e-07	8.195e-07	0.843

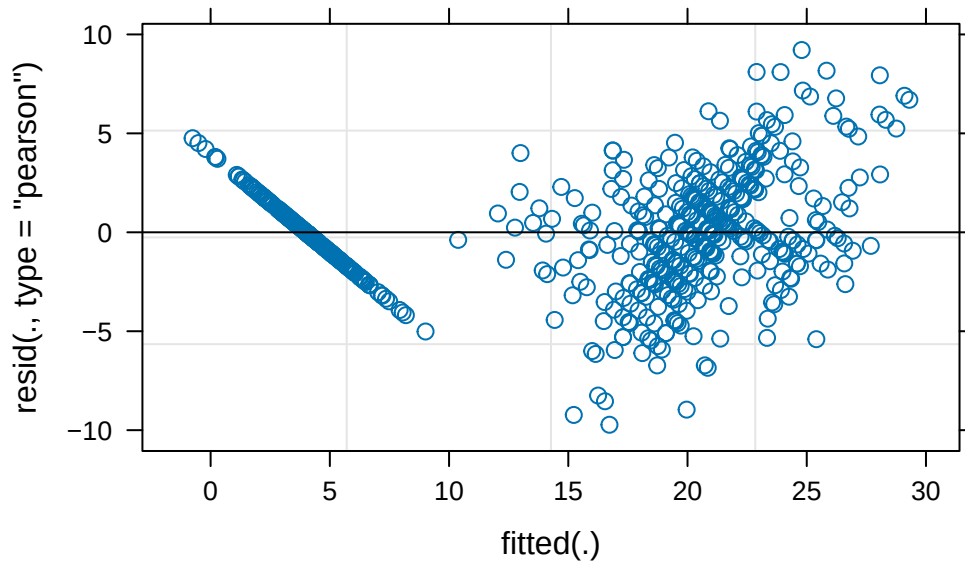
Correlation of Fixed Effects:

	(Intr)	age	I(g^2)	I(g^3)	sexmal	ag:sexm	I(^2):
age		-0.943					
I(age^2)		0.912	-0.991				
I(age^3)		-0.879	0.967	-0.992			
sexmale		-0.431	0.407	-0.394	0.379		
age:sexmale		0.381	-0.404	0.400	-0.391	-0.957	
I(g^2):sexml		-0.321	0.349	-0.352	0.349	0.925	-0.989
I(g^3):sexml		0.263	-0.289	0.296	-0.299	-0.886	0.961 -0.990

fit warnings:

Some predictor variables are on very different scales: consider rescaling

```
plot(mod_all_full)
```



```
confint(mod_all_full)
```

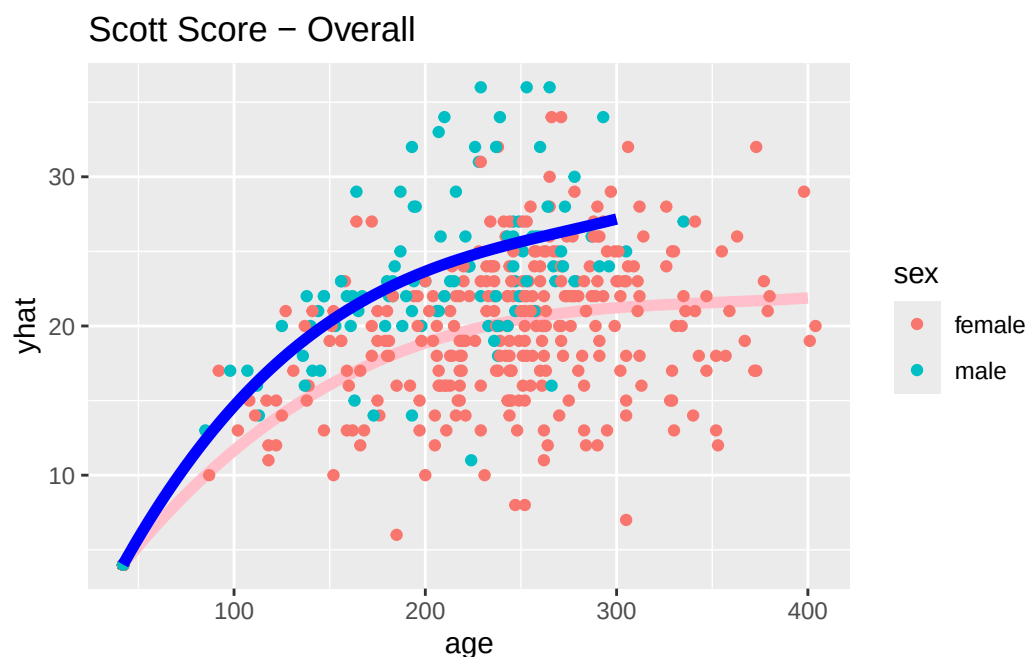
Computing profile confidence intervals ...

	2.5 %	97.5 %
.sig01	1.774347e+00	2.516106e+00
.sigma	2.828582e+00	3.250379e+00
(Intercept)	-5.361139e+00	-1.664398e+00
age	1.516552e-01	2.523818e-01
I(age^2)	-8.501675e-04	-2.603172e-04
I(age^3)	4.124907e-08	1.003273e-06
sexmale	-7.938213e+00	6.156152e-01
age:sexmale	-2.001948e-02	2.286823e-01
I(age^2):sexmale	-1.284882e-03	3.883343e-04
I(age^3):sexmale	-9.184339e-07	2.300077e-06

```
newx <- data.frame(age = 42:400, sex = "female")
newx$yhat <- (predict(mod_all_full, newdata = newx, re.form = ~0))
```

```
newxm <- data.frame(age = 42:300, sex = "male")
newxm$yhat <- (predict(mod_all_full, newdata = newxm, re.form = ~0))
```

```
ggplot(aes(x = age, y = yhat), data = newx) + geom_path(col = "pink", lwd = 2) + geom_point(
```



BM model

```
library(lme4)
mod_bm <- lmer(value ~ (age+I(age^2) + I(age^3))*sex + (1|id), data = scott_long %>% filter(BI == 1))
```

Warning: Some predictor variables are on very different scales: consider rescaling

```
mod_bm_red <- lmer(value ~ (age+I(age^2)+ I(age^3)) + (1|id), data = scott_long %>% filter(BI == 1))
```

Warning: Some predictor variables are on very different scales: consider rescaling

```
#ICC BM
(VarCorr(mod_bm)$id[1])/(VarCorr(mod_bm)$id[1]+attr(VarCorr(mod_bm),"sc")^2)
```

```
[1] 0.3120465
```

```
#test stat
-2*(logLik(mod_bm_red) - logLik(mod_bm))
```

```
'log Lik.' 27.50309 (df=6)
```

```
#p-value
1 - pchisq(-2*(logLik(mod_bm_red) - logLik(mod_bm)),4)
```

```
'log Lik.' 1.572599e-05 (df=6)
```

```
summary(mod_bm)
```

```
Linear mixed model fit by maximum likelihood ['lmerMod']
Formula: value ~ (age + I(age^2) + I(age^3)) * sex + (1 | id)
Data: scott_long %>% filter(BLMD == "BM")
```

AIC	BIC	logLik	deviance	df.resid
1397.2	1441.2	-688.6	1377.2	593

Scaled residuals:

Min	1Q	Median	3Q	Max
-4.8569	-0.4545	-0.0533	0.4862	3.1491

Random effects:

Groups	Name	Variance	Std.Dev.
id	(Intercept)	0.1958	0.4424
	Residual	0.4316	0.6570

Number of obs: 603, groups: id, 201

Fixed effects:

	Estimate	Std. Error	t value
(Intercept)	-2.266e+00	2.021e-01	-11.209
age	9.115e-02	5.502e-03	16.567
I(age^2)	-3.248e-04	3.223e-05	-10.079
I(age^3)	3.857e-07	5.259e-08	7.334
sexmale	-1.232e+00	4.690e-01	-2.627
age:sexmale	3.834e-02	1.363e-02	2.813
I(age^2):sexmale	-2.418e-04	9.168e-05	-2.638
I(age^3):sexmale	4.555e-07	1.764e-07	2.583

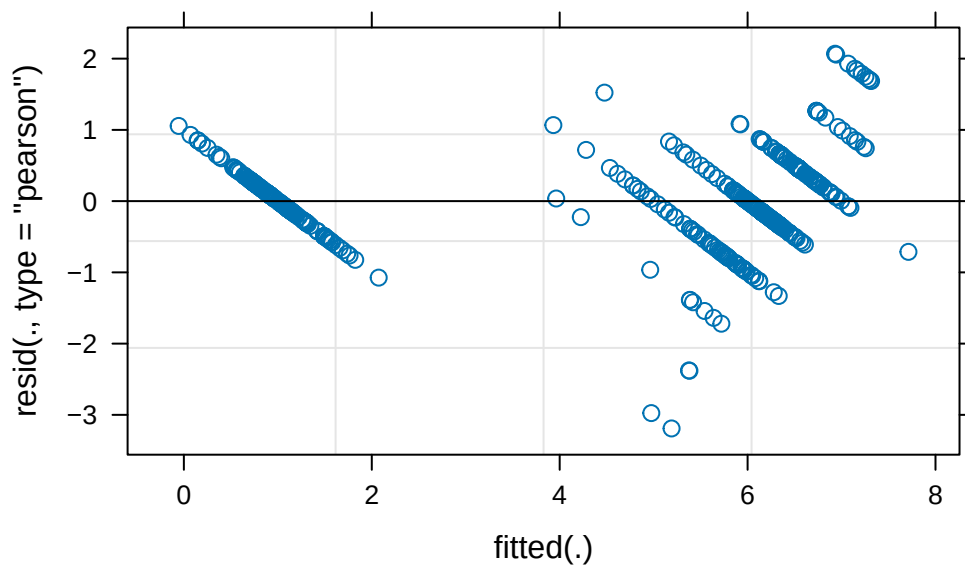
Correlation of Fixed Effects:

	(Intr)	age	I(g^2)	I(g^3)	sexmal	ag:sexm	I(^2):
age		-0.944					
I(age^2)		0.913	-0.991				
I(age^3)		-0.879	0.967	-0.992			
sexmale		-0.431	0.407	-0.393	0.379		
age:sexmale		0.381	-0.404	0.400	-0.390	-0.957	
I(g^2):sxm1		-0.321	0.348	-0.352	0.349	0.925	-0.989
I(g^3):sxm1		0.262	-0.288	0.296	-0.298	-0.886	0.961

fit warnings:

Some predictor variables are on very different scales: consider rescaling

```
plot(mod_bm)
```



```
confint(mod_bm)
```

Computing profile confidence intervals ...

Warning in profile.merMod(object, which = parm, signames = oldNames, ...):
non-monotonic profile for I(age^3)

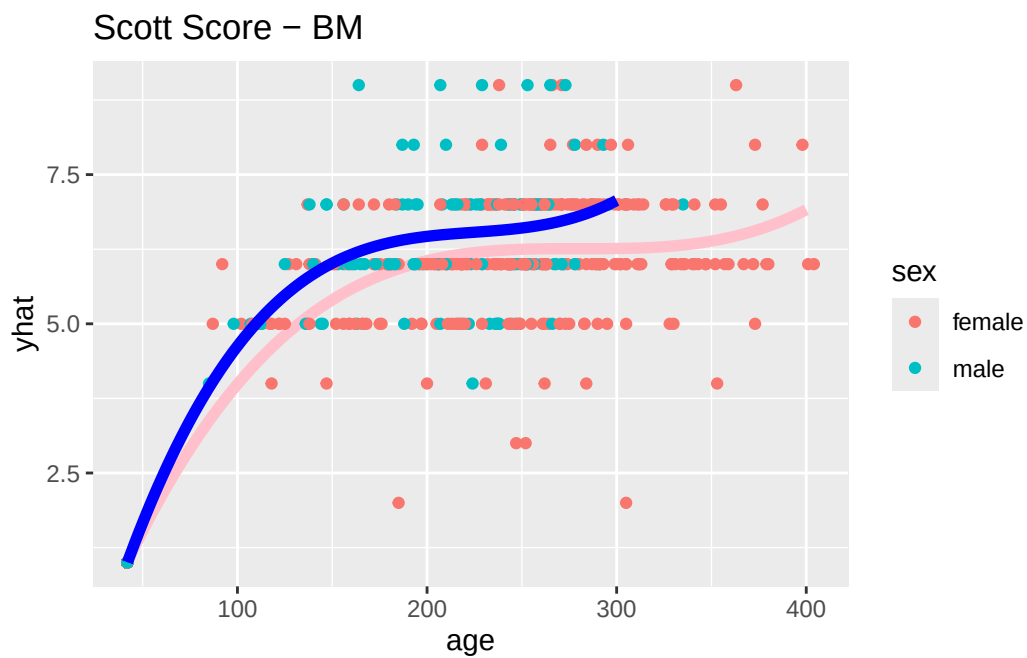
Warning in confint.thpr(pp, level = level, zeta = zeta): bad spline fit for
I(age^3): falling back to linear interpolation

	2.5 %	97.5 %
.sig01	3.641796e-01	5.241933e-01
.sigma	6.138803e-01	7.053054e-01
(Intercept)	-2.663009e+00	-1.868469e+00
age	8.033269e-02	1.019677e-01
I(age^2)	-3.881395e-04	-2.614562e-04
I(age^3)	2.823800e-07	4.890441e-07
sexmale	-2.152828e+00	-3.115144e-01
age:sexmale	1.158228e-02	6.509252e-02
I(age^2):sexmale	-4.217848e-04	-6.183189e-05
I(age^3):sexmale	1.092840e-07	8.017311e-07

```
newx <- data.frame(age = 42:400, sex = "female")
newx$yhat <- (predict(mod_bm, newdata = newx, re.form = ~0))

newxm <- data.frame(age = 42:300, sex = "male")
newxm$yhat <- (predict(mod_bm, newdata = newxm, re.form = ~0))

ggplot(aes(x = age, y = yhat), data = newx) + geom_path(col = "pink", lwd = 2) + geom_point(
```



BD Model

```
library(lme4)
mod_bd <- lmer(value ~ (age+I(age^2)+ I(age^3))*sex + (1|id), data = scott_long %>% filter(BLMD == "BD"))
```

Warning: Some predictor variables are on very different scales: consider rescaling

```
mod_bd_red <- lmer(value ~ (age+I(age^2)+ I(age^3)) + (1|id), data = scott_long %>% filter(BLMD == "BD"))
```

Warning: Some predictor variables are on very different scales: consider rescaling

```
#ICC BD
(VarCorr(mod_bd)$id[1])/(VarCorr(mod_bd)$id[1]+attr(VarCorr(mod_bd),"sc")^2)
```

```
[1] 0.3302695
```

```
#Test stat
-2*(logLik(mod_bd_red) - logLik(mod_bd))
```

```
'log Lik.' 50.48236 (df=6)
```

```
#p-value
1 - pchisq(-2*(logLik(mod_bd_red) - logLik(mod_bd)),4)
```

```
'log Lik.' 2.863381e-10 (df=6)
```

```
summary(mod_bd)
```

```
Linear mixed model fit by maximum likelihood ['lmerMod']
Formula: value ~ (age + I(age^2) + I(age^3)) * sex + (1 | id)
Data: scott_long %>% filter(BLMD == "BD")

      AIC      BIC  logLik deviance df.resid 
1483.8   1527.8   -731.9   1463.8     593   
```

Scaled residuals:

	Min	1Q	Median	3Q	Max
	-4.1758	-0.4768	0.0364	0.4429	2.7227

Random effects:

Groups	Name	Variance	Std.Dev.
id	(Intercept)	0.2417	0.4916
Residual		0.4901	0.7001

Number of obs: 603, groups: id, 201

Fixed effects:

	Estimate	Std. Error	t value
(Intercept)	-1.591e+00	2.171e-01	-7.328
age	7.152e-02	5.913e-03	12.096
I(age^2)	-2.417e-04	3.465e-05	-6.975
I(age^3)	2.879e-07	5.656e-08	5.090
sexmale	-6.894e-01	5.034e-01	-1.370
age:sexmale	2.054e-02	1.463e-02	1.404
I(age^2):sexmale	-1.078e-04	9.845e-05	-1.095
I(age^3):sexmale	1.982e-07	1.894e-07	1.046

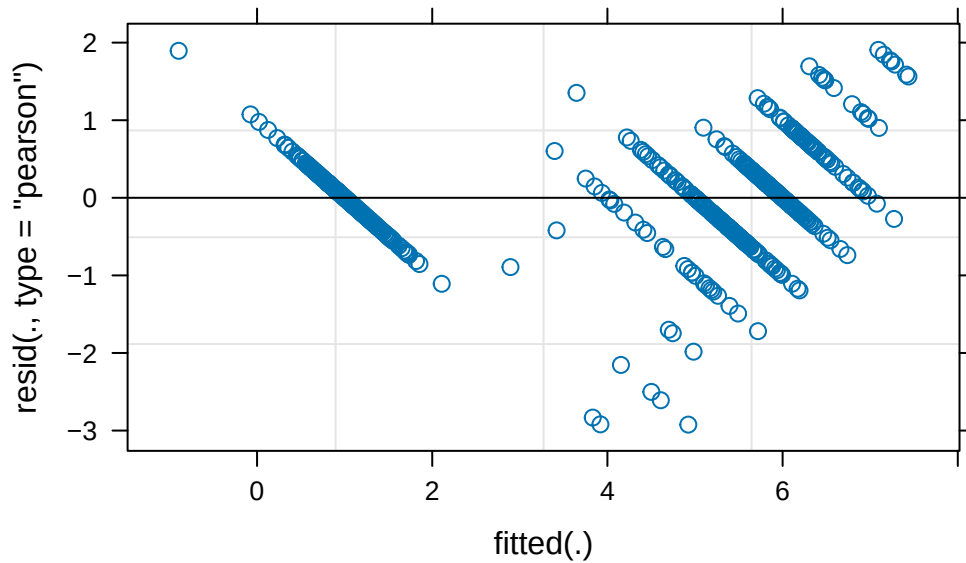
Correlation of Fixed Effects:

	(Intr)	age	I(g^2)	I(g^3)	sexmal	ag:sexm	I(^2):
age		-0.944					
I(age^2)		0.913	-0.991				
I(age^3)		-0.879	0.967	-0.992			
sexmale		-0.431	0.407	-0.394	0.379		
age:sexmale		0.381	-0.404	0.400	-0.391	-0.957	
I(g^2):sexml		-0.321	0.349	-0.352	0.349	0.925	-0.989
I(g^3):sexml		0.263	-0.289	0.296	-0.299	-0.886	0.961

fit warnings:

Some predictor variables are on very different scales: consider rescaling

```
plot(mod_bd)
```



```
confint(mod_bm)
```

Computing profile confidence intervals ...

Warning in profile.merMod(object, which = parm, signames = oldNames, ...):
non-monotonic profile for I(age^3)

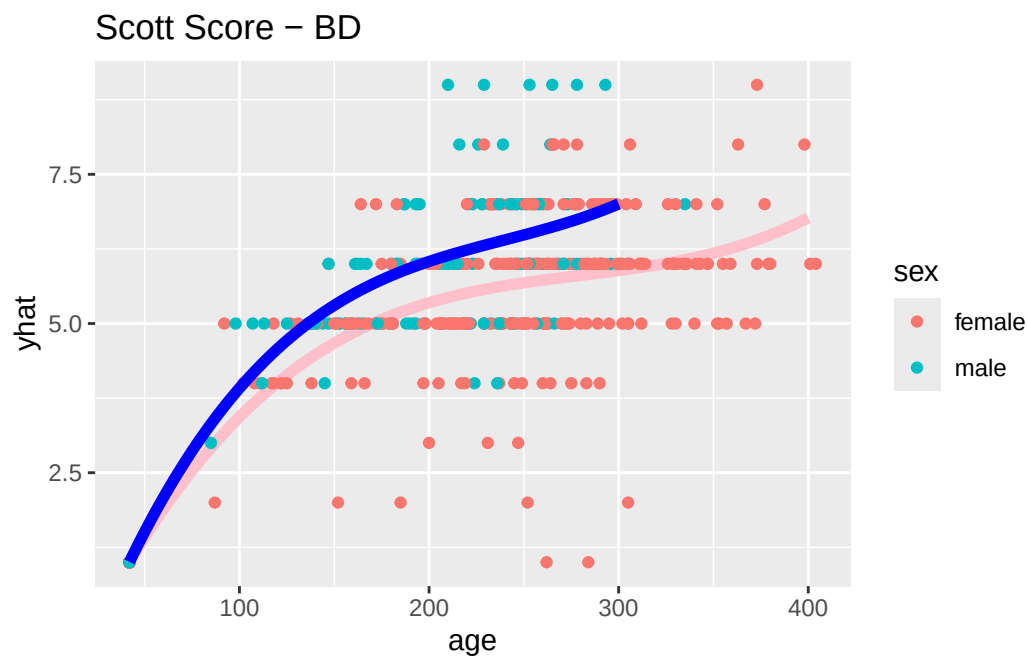
Warning in confint.thpr(pp, level = level, zeta = zeta): bad spline fit for
I(age^3): falling back to linear interpolation

	2.5 %	97.5 %
.sig01	3.641796e-01	5.241933e-01
.sigma	6.138803e-01	7.053054e-01
(Intercept)	-2.663009e+00	-1.868469e+00
age	8.033269e-02	1.019677e-01
I(age^2)	-3.881395e-04	-2.614562e-04
I(age^3)	2.823800e-07	4.890441e-07
sexmale	-2.152828e+00	-3.115144e-01
age:sexmale	1.158228e-02	6.509252e-02
I(age^2):sexmale	-4.217848e-04	-6.183189e-05
I(age^3):sexmale	1.092840e-07	8.017311e-07

```
newx <- data.frame(age = 42:400, sex = "female")
newx$yhat <- (predict(mod_bd, newdata = newx, re.form = ~0))

newxm <- data.frame(age = 42:300, sex = "male")
newxm$yhat <- (predict(mod_bd, newdata = newxm, re.form = ~0))

ggplot(aes(x = age, y = yhat), data = newx) + geom_path(col = "pink", lwd = 2) + geom_point(col = "pink", lwd = 2) +
```



LM model

```
library(lme4)
mod_lm <- lmer(value ~ (age+I(age^2)+ I(age^3))*sex + (1|id), data = scott_long %>% filter(BL == "BD"))
```

Warning: Some predictor variables are on very different scales: consider rescaling

```
mod_lm_red <- lmer(value ~ (age+I(age^2)+ I(age^3)) + (1|id), data = scott_long %>% filter(BL == "BD"))
```

Warning: Some predictor variables are on very different scales: consider rescaling

```
#ICC LM
(VarCorr(mod_lm)$id[1])/(VarCorr(mod_lm)$id[1]+attr(VarCorr(mod_lm),"sc")^2)
```

```
[1] 0.2978919
```

```
#test stat
-2*(logLik(mod_lm_red) - logLik(mod_lm))
```

```
'log Lik.' 80.88223 (df=6)
```

```
#p-value
1 - pchisq(-2*(logLik(mod_lm_red) - logLik(mod_lm)),4)
```

```
'log Lik.' 1.110223e-16 (df=6)
```

```
summary(mod_lm)
```

```
Linear mixed model fit by maximum likelihood ['lmerMod']
Formula: value ~ (age + I(age^2) + I(age^3)) * sex + (1 | id)
Data: scott_long %>% filter(BLMD == "LM")
```

AIC	BIC	logLik	deviance	df.resid
2070.1	2114.1	-1025.0	2050.1	593

Scaled residuals:

Min	1Q	Median	3Q	Max
-2.71290	-0.46049	0.02788	0.46279	2.86228

Random effects:

Groups	Name	Variance	Std.Dev.
id	(Intercept)	0.566	0.7523
Residual		1.334	1.1550

Number of obs: 603, groups: id, 201

Fixed effects:

	Estimate	Std. Error	t value
(Intercept)	-1.146e-01	3.532e-01	-0.324
age	2.787e-02	9.608e-03	2.901
I(age^2)	-2.609e-05	5.626e-05	-0.464

I(age ³)	-3.895e-08	9.180e-08	-0.424
sexmale	-6.130e-01	8.199e-01	-0.748
age:sexmale	1.524e-02	2.382e-02	0.640
I(age ²):sexmale	-1.717e-05	1.602e-04	-0.107
I(age ³):sexmale	-2.121e-08	3.082e-07	-0.069

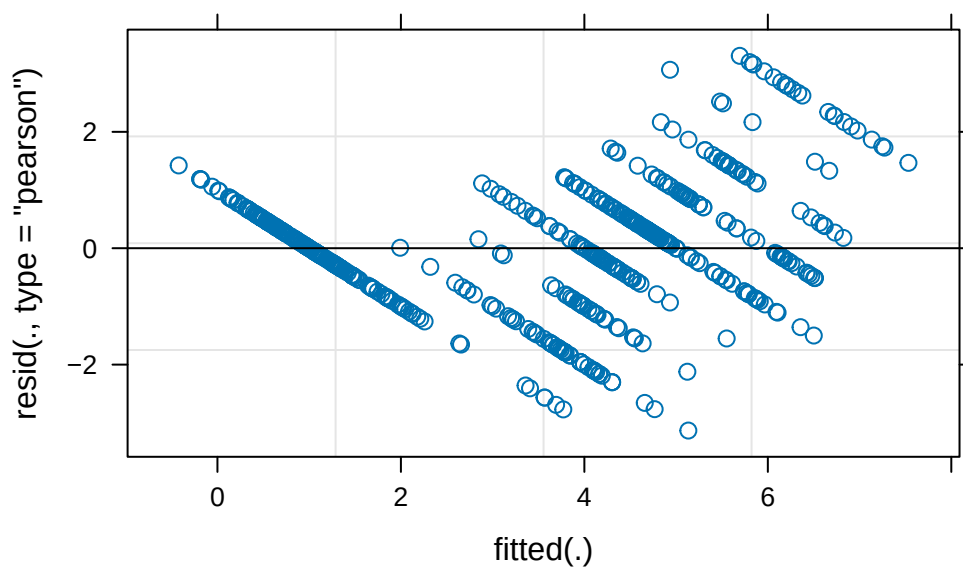
Correlation of Fixed Effects:

	(Intr)	age	I(g ²)	I(g ³)	sexmal	ag:sexm	I(²):
age	-0.945						
I(age ²)	0.913	-0.991					
I(age ³)	-0.879	0.967	-0.992				
sexmale	-0.431	0.407	-0.393	0.379			
age:sexmale	0.381	-0.403	0.400	-0.390	-0.958		
I(g ²):sexml	-0.321	0.348	-0.351	0.348	0.925	-0.989	
I(g ³):sexml	0.262	-0.288	0.295	-0.298	-0.886	0.961	-0.990

fit warnings:

Some predictor variables are on very different scales: consider rescaling

```
plot(mod_lm)
```



```
confint(mod_lm)
```

Computing profile confidence intervals ...

2.5 %	97.5 %
-------	--------


```

.sig01          6.138133e-01 8.956860e-01
.sigma          1.079227e+00 1.240107e+00
(Intercept)     -8.098491e-01 5.806864e-01
age             8.940271e-03 4.679066e-02
I(age^2)        -1.367742e-04 8.468194e-05
I(age^3)        -2.195181e-07 1.414884e-07
sexmale         -2.222479e+00 9.966918e-01
age:sexmale     -3.151813e-02 6.200393e-02
I(age^2):sexmale -3.316920e-04 2.973565e-04
I(age^3):sexmale -6.263407e-07 5.838899e-07

```

```

newx <- data.frame(age = 1:400, sex = "female")
newx$yhat <- (predict(mod_lm, newdata = newx, re.form = ~0))

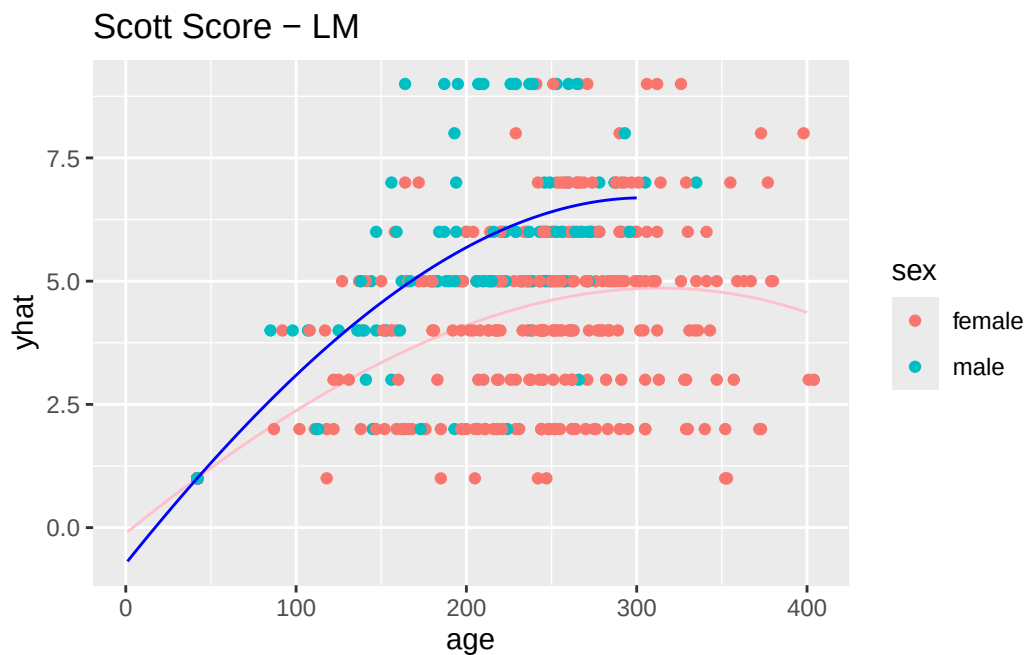
newxm <- data.frame(age = 1:300, sex = "male")
newxm$yhat <- (predict(mod_lm, newdata = newxm, re.form = ~0))

```

```

ggplot(aes(x = age, y = yhat), data = newx) + geom_path(col = "pink") + geom_point(aes(x = age, y = yhat, sex = sex), col = "pink")

```



LD Model

```
library(lme4)
mod_ld <- lmer(value ~ (age+I(age^2)+ I(age^3))*sex + (1|id), data = scott_long %>% filter(BLMD == "LD")
```

Warning: Some predictor variables are on very different scales: consider rescaling

```
mod_ld_red <- lmer(value ~ (age+I(age^2)+ I(age^3)) + (1|id), data = scott_long %>% filter(BLMD == "LD")
```

Warning: Some predictor variables are on very different scales: consider rescaling

```
#ICC LD
(VarCorr(mod_ld)$id[1])/(VarCorr(mod_ld)$id[1]+attr(VarCorr(mod_ld),"sc")^2)
```

```
[1] 0.3381656
```

```
#Test stat
-2*(logLik(mod_ld_red) - logLik(mod_ld))
```

```
'log Lik.' 136.2135 (df=6)
```

```
#p-value
1 - pchisq(-2*(logLik(mod_ld_red) - logLik(mod_ld)),4)
```

```
'log Lik.' 0 (df=6)
```

```
summary(mod_ld)
```

```
Linear mixed model fit by maximum likelihood ['lmerMod']
Formula: value ~ (age + I(age^2) + I(age^3)) * sex + (1 | id)
Data: scott_long %>% filter(BLMD == "LD")
```

AIC	BIC	logLik	deviance	df.resid
2024.8	2068.8	-1002.4	2004.8	593

Scaled residuals:

Min	1Q	Median	3Q	Max
-----	----	--------	----	-----

-2.93811 -0.53078 0.01785 0.54850 2.92456

Random effects:

Groups	Name	Variance	Std.Dev.
id	(Intercept)	0.6099	0.781
Residual		1.1937	1.093

Number of obs: 603, groups: id, 201

Fixed effects:

	Estimate	Std. Error	t value
(Intercept)	4.364e-01	3.400e-01	1.284
age	1.216e-02	9.261e-03	1.313
I(age^2)	3.375e-05	5.428e-05	0.622
I(age^3)	-1.077e-07	8.860e-08	-1.215
sexmale	-1.121e+00	7.880e-01	-1.423
age:sexmale	3.002e-02	2.291e-02	1.310
I(age^2):sexmale	-8.021e-05	1.541e-04	-0.520
I(age^3):sexmale	5.577e-08	2.965e-07	0.188

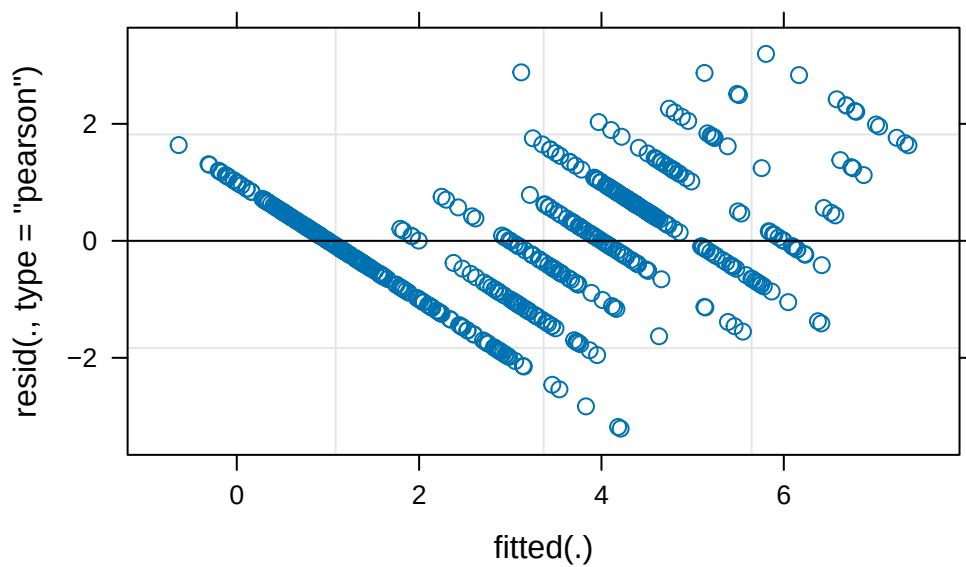
Correlation of Fixed Effects:

	(Intr)	age	I(g^2)	I(g^3)	sexmal	ag:sexm	I(^2):
age	-0.943						
I(age^2)	0.912	-0.991					
I(age^3)	-0.879	0.967	-0.992				
sexmale	-0.431	0.407	-0.394	0.379			
age:sexmale	0.381	-0.404	0.401	-0.391	-0.957		
I(g^2):sexml	-0.321	0.349	-0.352	0.349	0.925	-0.989	
I(g^3):sexml	0.263	-0.289	0.296	-0.299	-0.886	0.961	-0.990

fit warnings:

Some predictor variables are on very different scales: consider rescaling

```
plot(mod_ld)
```



```
confint(mod_bm)
```

Computing profile confidence intervals ...

Warning in profile.merMod(object, which = parm, signames = oldNames, ...):
non-monotonic profile for I(age^3)

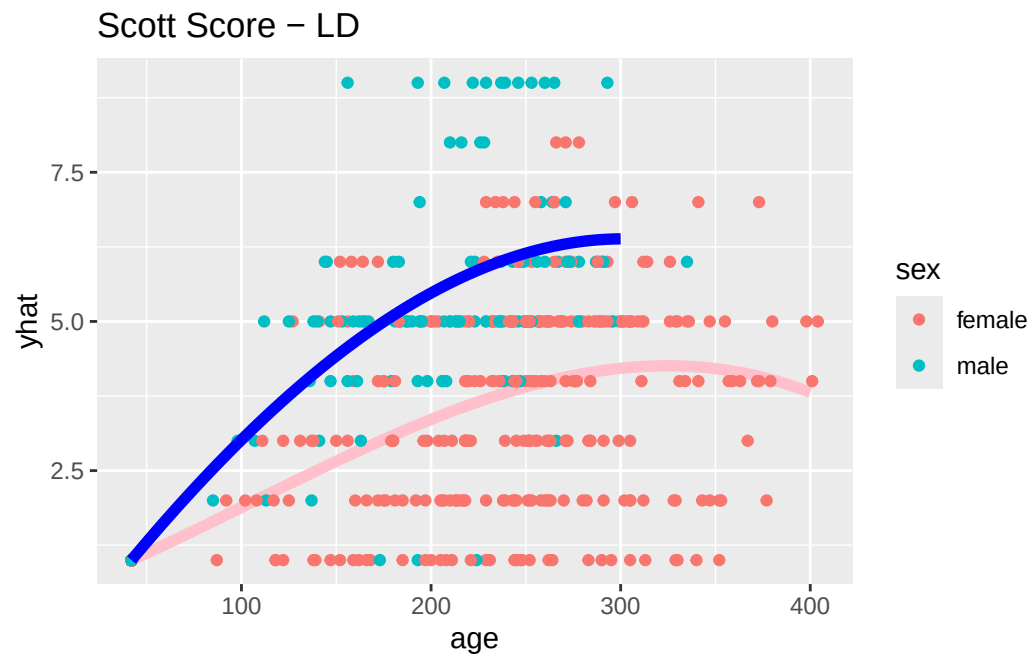
Warning in confint.thpr(pp, level = level, zeta = zeta): bad spline fit for
I(age^3): falling back to linear interpolation

	2.5 %	97.5 %
.sig01	3.641796e-01	5.241933e-01
.sigma	6.138803e-01	7.053054e-01
(Intercept)	-2.663009e+00	-1.868469e+00
age	8.033269e-02	1.019677e-01
I(age^2)	-3.881395e-04	-2.614562e-04
I(age^3)	2.823800e-07	4.890441e-07
sexmale	-2.152828e+00	-3.115144e-01
age:sexmale	1.158228e-02	6.509252e-02
I(age^2):sexmale	-4.217848e-04	-6.183189e-05
I(age^3):sexmale	1.092840e-07	8.017311e-07

```
newx <- data.frame(age = 42:400, sex = "female")
newx$yhat <- (predict(mod_ld, newdata = newx, re.form = ~0))

newxm <- data.frame(age = 42:300, sex = "male")
newxm$yhat <- (predict(mod_ld, newdata = newxm, re.form = ~0))

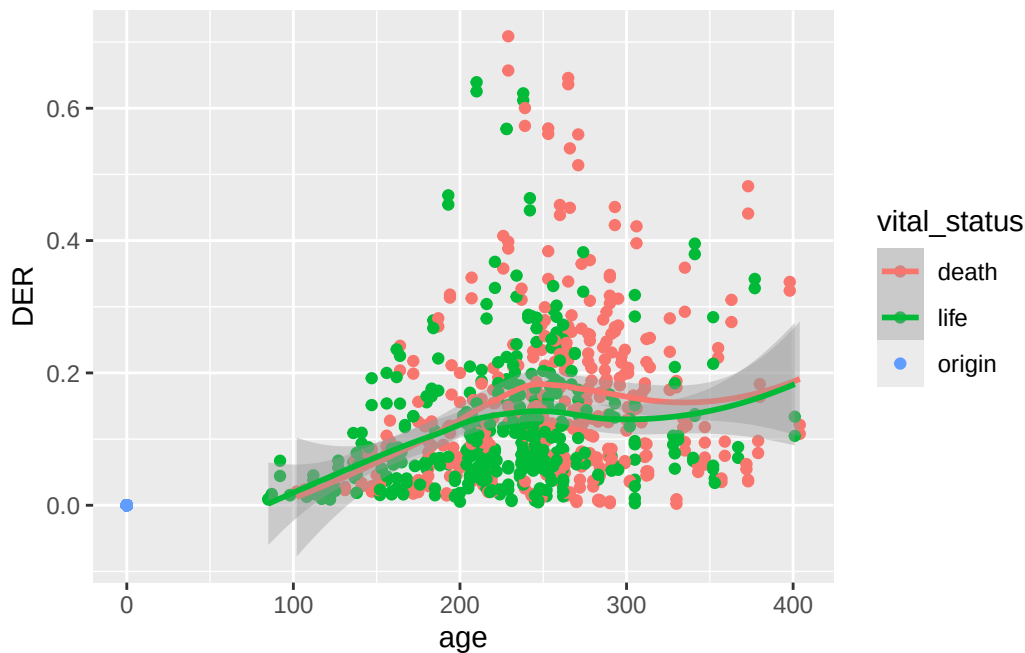
ggplot(aes(x = age, y = yhat), data = newx) + geom_path(col = "pink", lwd = 2) + geom_point()
```



Dentin Ratio

```
ggplot(aes(x = age, y = DER, color = vital_status), data = dentin_long) + geom_point() + geom
```

``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



```
library(lme4)
mod_dentin_full <- lmer(DER ~ (age+I(age^2)+ I(age^3))*sex + (1|id) + (1|rater), data = dentin_long)
```

Warning: Some predictor variables are on very different scales: consider rescaling

```
mod_dentin_red <- lmer(DER ~ (age+I(age^2)+ I(age^3)) + (1|id) + (1|rater), data = dentin_long)
```

Warning: Some predictor variables are on very different scales: consider rescaling

```
#Summary stats for dentin
#By sex
dentin_long %>% group_by(id, vital_status, sex) %>% summarize(DER = mean(DER)) %>% ungroup()
```

`summarise()` has grouped output by 'id', 'vital_status'. You can override using the `.groups` argument.
 `summarise()` has grouped output by 'vital_status'. You can override using the `.groups` argument.

```
# A tibble: 6 x 4
# Groups:   vital_status [3]
  vital_status sex      mean    std
  <chr>        <chr> <dbl> <dbl>
1 death      female 0.132 0.104
2 death      male   0.206 0.163
3 life       female 0.106 0.0938
4 life       male   0.152 0.133
5 origin     female 0      0
6 origin     male   0      0
```

```
#Overall
dentin_long %>% group_by(id, vital_status, sex) %>% summarize(DER = mean(DER)) %>% ungroup()
```

`summarise()` has grouped output by 'id', 'vital_status'. You can override using the `.groups` argument.

```
# A tibble: 3 x 3
  vital_status mean    std
  <chr>        <dbl> <dbl>
1 death      0.151 0.126
2 life       0.118 0.107
3 origin     0      0
```

```
#comparing vital status
library(lme4)
mod <- lmer(DER ~ vital_status + (1|rater) + (1|id), data = dentin_long %>% filter(vital_status != "origin"))

#Test stat
-2*(logLik(mod_dentin_red) - logLik(mod_dentin_full))
```

```
'log Lik.' 137.6153 (df=7)
```

```
#p-value
1 - pchisq(-2*(logLik(mod_dentin_red) - logLik(mod_dentin_full)),4)
```

```
'log Lik.' 0 (df=7)
```

```
summary(mod_dentin_full)
```

Linear mixed model fit by maximum likelihood ['lmerMod']

Formula: DER ~ (age + I(age^2) + I(age^3)) * sex + (1 | id) + (1 | rater)

Data: dentin_long

AIC	BIC	logLik	deviance	df.resid
-2974.3	-2918.2	1498.1	-2996.3	1195

Scaled residuals:

Min	1Q	Median	3Q	Max
-4.6371	-0.5037	-0.0729	0.4644	4.0732

Random effects:

Groups	Name	Variance	Std.Dev.
id	(Intercept)	4.145e-03	0.064382
rater	(Intercept)	1.709e-05	0.004134
Residual		3.429e-03	0.058556

Number of obs: 1206, groups: id, 201; rater, 2

Fixed effects:

	Estimate	Std. Error	t value
(Intercept)	-3.198e-05	6.899e-03	-0.005
age	-2.793e-04	2.329e-04	-1.199
I(age^2)	5.667e-06	1.692e-06	3.350
I(age^3)	-1.009e-08	3.031e-09	-3.327
sexmale	-1.015e-04	1.241e-02	-0.008
age:sexmale	-5.171e-04	5.299e-04	-0.976
I(age^2):sexmale	7.898e-06	4.524e-06	1.746
I(age^3):sexmale	-1.632e-08	9.639e-09	-1.693

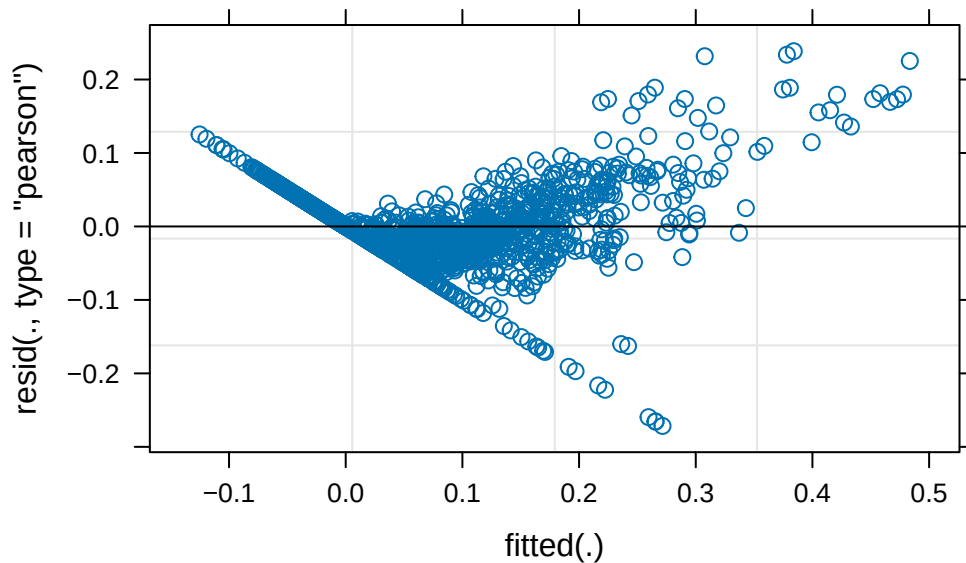
Correlation of Fixed Effects:

	(Intr)	age	I(g^2)	I(g^3)	sexmal	ag:sexm	I(^2):
age		-0.094					
I(age^2)	0.053	-0.986					
I(age^3)	-0.039	0.952	-0.989				
sexmale	-0.456	0.052	-0.030	0.021			
age:sexmale	0.041	-0.440	0.433	-0.419	-0.101		
I(g^2):sexml	-0.020	0.368	-0.374	0.370	0.055	-0.982	
I(g^3):sexml	0.012	-0.299	0.311	-0.314	-0.039	0.942	-0.988

fit warnings:

Some predictor variables are on very different scales: consider rescaling


```
plot(mod_dentin_full)
```



```
confint(mod_dentin_full)
```

Computing profile confidence intervals ...

Warning in profile.merMod(object, which = parm, signames = oldNames, ...):
non-monotonic profile for I(age^3)

Warning in profile.merMod(object, which = parm, signames = oldNames, ...):
non-monotonic profile for I(age^3):sexmale

Warning in confint.thpr(pp, level = level, zeta = zeta): bad spline fit for
I(age^3): falling back to linear interpolation

Warning in confint.thpr(pp, level = level, zeta = zeta): bad spline fit for
I(age^3):sexmale: falling back to linear interpolation

	2.5 %	97.5 %
.sig01	5.769599e-02	7.213582e-02
.sig02	0.000000e+00	2.170873e-02
.sigma	5.608466e-02	6.121426e-02
(Intercept)	-1.511011e-02	1.504710e-02

```

age                -7.361188e-04  1.775428e-04
I(age^2)           2.349205e-06  8.985359e-06
I(age^3)           -1.603070e-08 -4.139735e-09
sexmale            -2.449331e-02  2.429104e-02
age:sexmale        -1.556596e-03  5.223088e-04
I(age^2):sexmale   -9.762482e-07  1.677352e-05
I(age^3):sexmale   -3.523132e-08  2.590469e-09

```

```
#ICC on model with eruption time point
```

```
(VarCorr(mod_dentin_full)$rater[1])/(VarCorr(mod_dentin_full)$rater[1] + VarCorr(mod_dentin_full)$id[1])
```

```
[1] 0.002251748
```

```
(VarCorr(mod_dentin_full)$id[1])/(VarCorr(mod_dentin_full)$rater[1] + VarCorr(mod_dentin_full)$id[1])
```

```
[1] 0.5460536
```

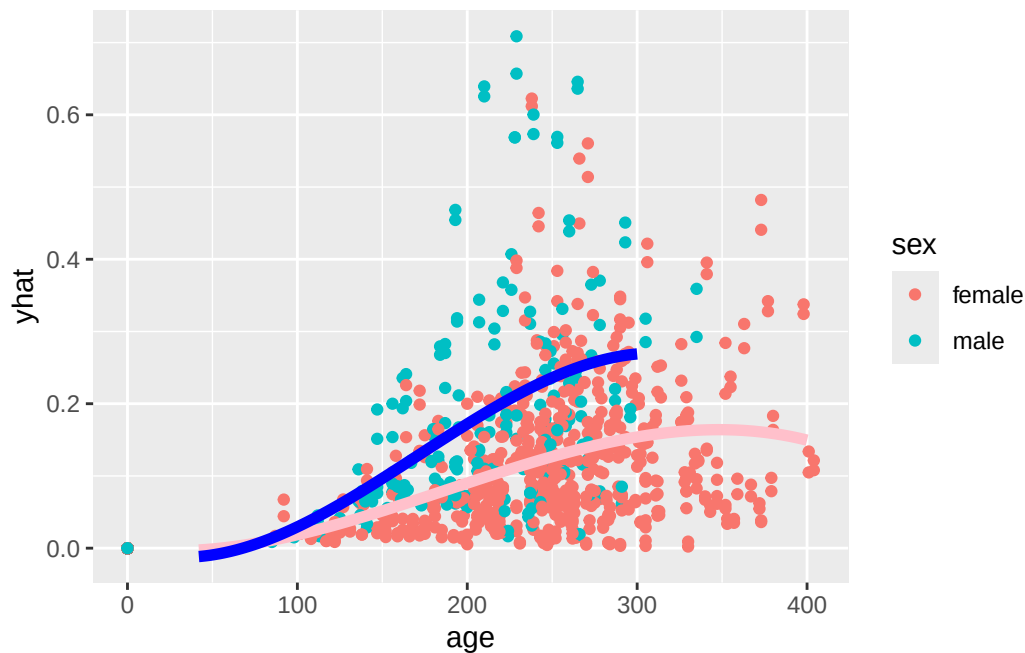
```

newx <- data.frame(age = 42:400, sex = "female")
newx$yhat <- (predict(mod_dentin_full, newdata = newx, re.form = ~0))

newxm <- data.frame(age = 42:300, sex = "male")
newxm$yhat <- (predict(mod_dentin_full, newdata = newxm, re.form = ~0))

ggplot(aes(x = age, y = yhat), data = newx) +
  geom_point(aes(x = age, y = DER, col = sex), data = dentin_long) + geom_path(col = "pink",

```



```
##ICC analysis
## Only use death and cast ages
mod_dentin_full <- lmer(DER ~ (age+I(age^2)+ I(age^3))*sex + (1|id) + (1|rater), data = dentin_lo
```

Warning: Some predictor variables are on very different scales: consider rescaling

```
mod_dentin_red <- lmer(DER ~ (age+I(age^2)+ I(age^3)) + (1|id) + (1|rater), data = dentin_lo
```

Warning: Some predictor variables are on very different scales: consider rescaling

```
(VarCorr(mod_dentin_full)$rater[1])/(VarCorr(mod_dentin_full)$rater[1] + VarCorr(mod_dentin_
```

```
[1] 0.004179212
```

```
(VarCorr(mod_dentin_full)$id[1])/(VarCorr(mod_dentin_full)$rater[1] + VarCorr(mod_dentin_ful
```

```
[1] 0.9222974
```

Are there any patterns for rate of wear? That is, is there a standard wear through time (by month or year), or are some baboons wearing their teeth down faster than what we might expect?

Pava

Dentin ratio

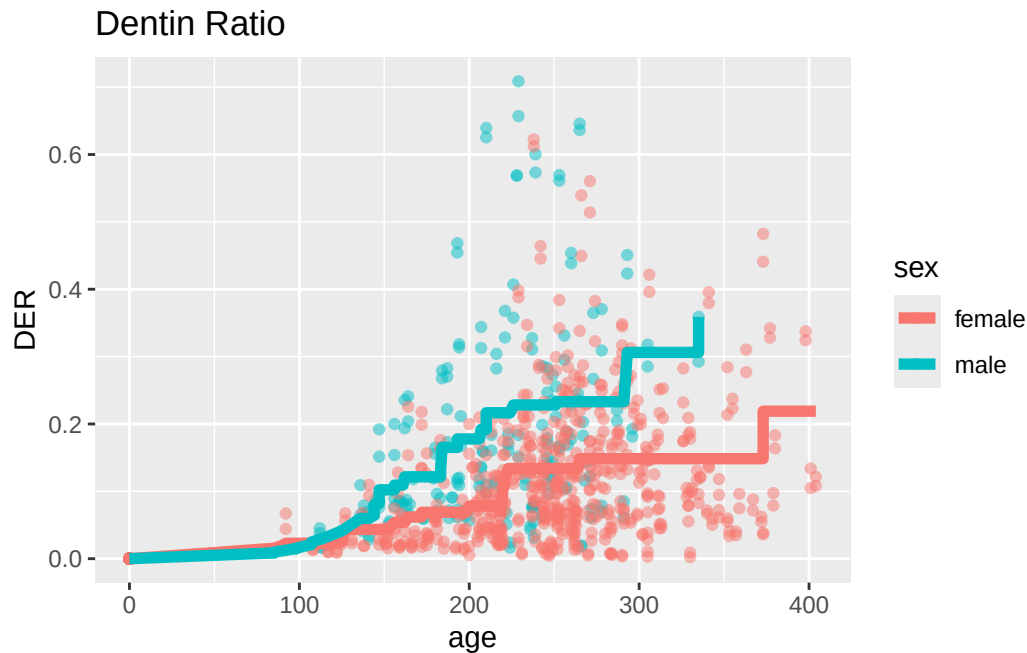
```
#Overall  
library(Iso)
```

Iso 0.0-21

An "infelicity" in the function `ufit()` (whereby it was all too easy to conflate the location of the mode with its index in the entries of the "x" argument) has been corrected. To this end, `ufit()` now has arguments "lmode" (the location of the mode), and "imode" (its index). At most one of these arguments should be specified. See the help for `ufit()`.

```
#Isotonic regression for BM.  
#Separate curves for male and female.  
sub <- dentin_long %>% filter(sex == "male") %>% arrange(age)  
y <- dentin_long %>% filter(sex == "male") %>% arrange(age) %>% pull(DER)  
sub$isoy <- pava(y)  
submale <- sub  
  
sub <- dentin_long %>% filter(sex == "female") %>% arrange(age)  
y <- dentin_long %>% filter(sex == "female") %>% arrange(age) %>% pull(DER)  
sub$isoy <- pava(y)  
subfemale <- sub  
  
sub_dentin <- rbind(submale, subfemale)
```

```
ggplot(aes(x = age, y = DER, color = sex), data = sub_dentin) + geom_point(alpha = 0.5) + ge
```



Scott scores

Scott scores summed across locations

```
#Overall
library(Iso)

#Isotonic regression for overall
scott_long_sum <- scott_long %>% group_by(id, castost, age, sex) %>% summarize(value = sum(v
```

``summarise()`` has grouped output by 'id', 'castost', 'age'. You can override using the ``.groups`` argument.

```
#Separate curves for male and female.
sub <- scott_long_sum %>% filter(sex == "male") %>% arrange(age)
y <- scott_long_sum %>% filter(sex == "male") %>% arrange(age) %>% pull(value)
sub$iso <- pava(y)
submale <- sub

sub <- scott_long_sum %>% filter(sex == "female") %>% arrange(age)
y <- scott_long_sum %>% filter(sex == "female") %>% arrange(age) %>% pull(value)
```

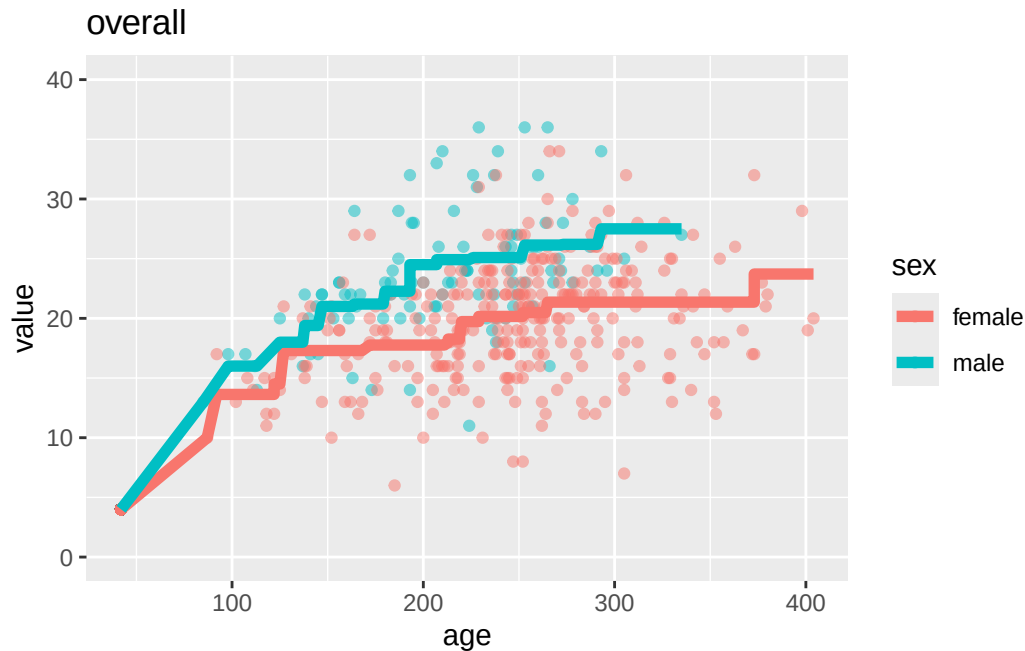
```

sub$iso_y <- pava(y)
subfemale <- sub

suball <- rbind(submale, subfemale)

ggplot(aes(x = age, y = value, color = sex), data = suball) + geom_point(alpha = 0.5) + geom_

```



Individual locations

```

library(Iso)

#Isotonic regression for BM.
#Separate curves for male and female.
type <- "BM"
sub <- scott_long %>% filter(BLMD == type & sex == "male") %>% arrange(age)
y <- scott_long %>% filter(BLMD == type & sex == "male") %>% arrange(age) %>% pull(value)
sub$iso_y <- pava(y)
submale <- sub

sub <- scott_long %>% filter(BLMD == type & sex == "female") %>% arrange(age)
y <- scott_long %>% filter(BLMD == type & sex == "female") %>% arrange(age) %>% pull(value)

```

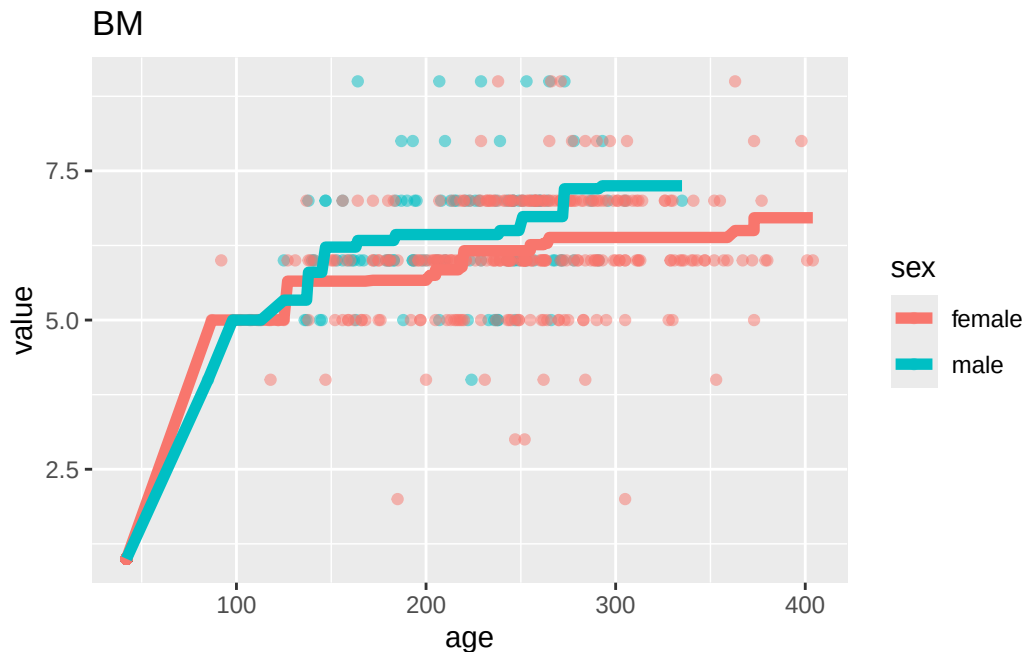
```

sub$isoy <- pava(y)
subfemale <- sub

subBM <- rbind(submale, subfemale)

ggplot(aes(x = age, y = value, color = sex), data = subBM) + geom_point(alpha = 0.5) + geom_l

```



```

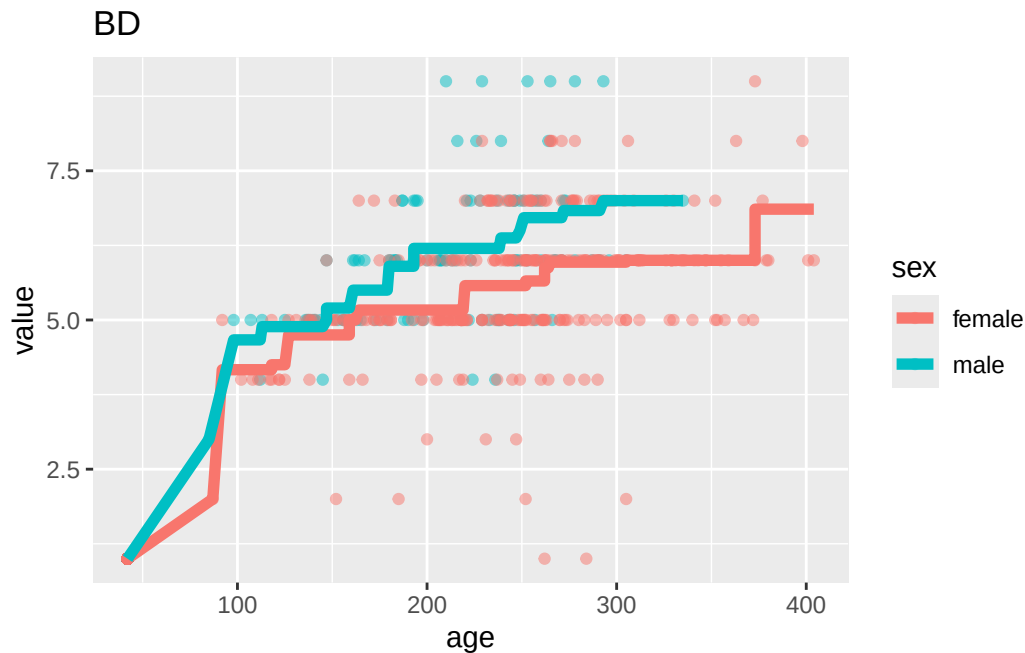
#Isotonic regression for BD.
#Separate curves for male and female.
type <- "BD"
sub <- scott_long %>% filter(BLMD == type & sex == "male") %>% arrange(age)
y <- scott_long %>% filter(BLMD == type & sex == "male") %>% arrange(age) %>% pull(value)
sub$isoy <- pava(y)
submale <- sub

sub <- scott_long %>% filter(BLMD == type & sex == "female") %>% arrange(age)
y <- scott_long %>% filter(BLMD == type & sex == "female") %>% arrange(age) %>% pull(value)
sub$isoy <- pava(y)
subfemale <- sub

subBD <- rbind(submale, subfemale)

ggplot(aes(x = age, y = value, color = sex), data = subBD) + geom_point(alpha = 0.5) + geom_l

```

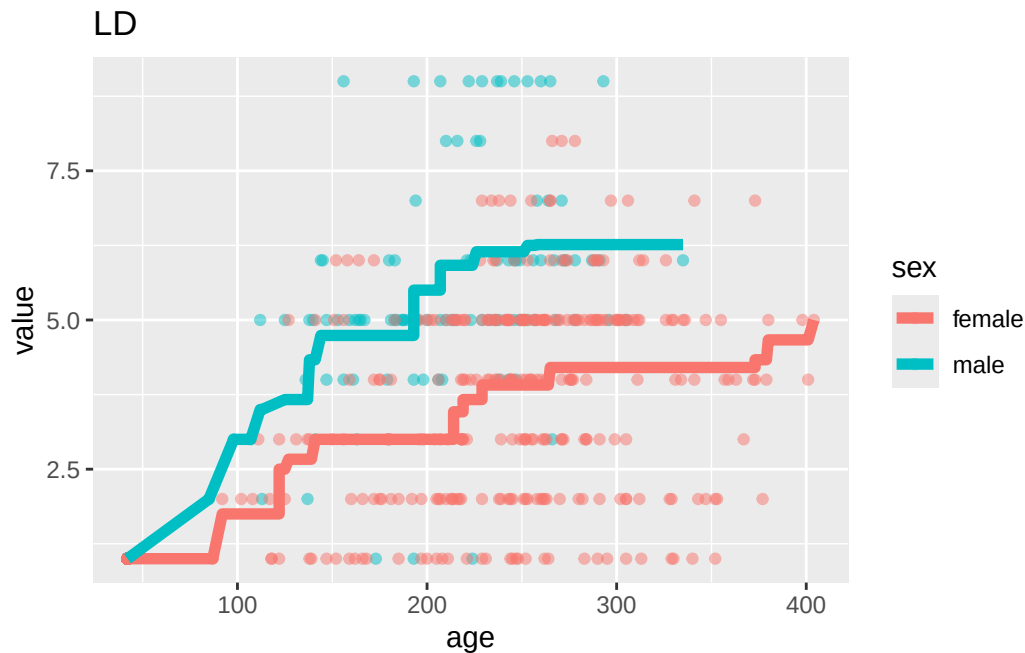


```
#Isotonic regression for LD.
#Separate curves for male and female.
type <- "LD"
sub <- scott_long %>% filter(BLMD == type & sex == "male") %>% arrange(age)
y <- scott_long %>% filter(BLMD == type & sex == "male") %>% arrange(age) %>% pull(value)
sub$isoy <- pava(y)
submale <- sub

sub <- scott_long %>% filter(BLMD == type & sex == "female") %>% arrange(age)
y <- scott_long %>% filter(BLMD == type & sex == "female") %>% arrange(age) %>% pull(value)
sub$isoy <- pava(y)
subfemale <- sub

subLD <- rbind(submale, subfemale)

ggplot(aes(x = age, y = value, color = sex), data = subLD) + geom_point(alpha = 0.5) + geom_
```

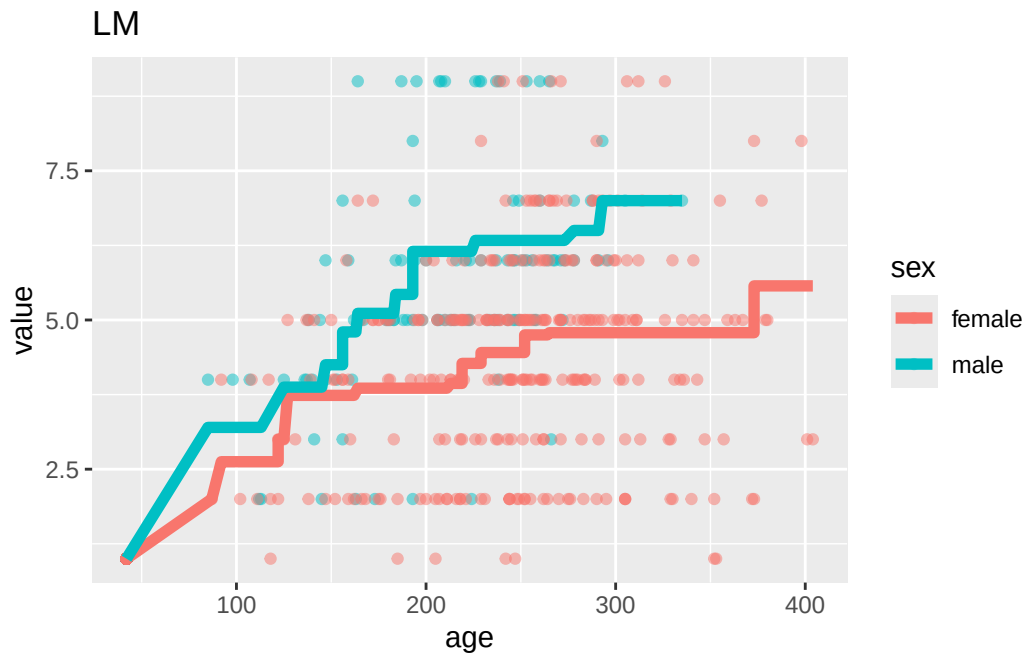



```
#Isotonic regression for LM.
#Separate curves for male and female.
type <- "LM"
sub <- scott_long %>% filter(BLMD == type & sex == "male") %>% arrange(age)
y <- scott_long %>% filter(BLMD == type & sex == "male") %>% arrange(age) %>% pull(value)
sub$isoy <- pava(y)
submale <- sub

sub <- scott_long %>% filter(BLMD == type & sex == "female") %>% arrange(age)
y <- scott_long %>% filter(BLMD == type & sex == "female") %>% arrange(age) %>% pull(value)
sub$isoy <- pava(y)
subfemale <- sub

subLM <- rbind(submale, subfemale)

ggplot(aes(x = age, y = value, color = sex), data = subLM) + geom_point(alpha = 0.5) + geom_
```



```
suball <- rbind(subBM, subBD, subLM, subLD)

# ggplot() + geom_point(aes(y = value, x = age, color = sex),data = suball,alpha = 0.5) + geom_

suball <- suball %>% mutate(BLword = case_when(BL == "B" ~ "Buccal",
                                              BL == "L" ~ "Lingual"),
                          MDword = case_when(MD == "M" ~ "Mesial",
                                              MD == "D" ~ "Distal"))

pdf(file = "/Users/gregorymatthews/Dropbox/baboon_git/isotonic.pdf", h = 5, w = 5, pointsize
ggplot() + geom_point(aes(y = value, x = age, color = sex),data = suball, size = 0.5) + geom_
  values = c("#F8766D","#00BFC4"),
  labels = c( "Female","Male")
)
dev.off()
```

```
pdf
2
```

```
pdf(file = "/Users/gregorymatthews/Dropbox/baboon_git/isotonic_qword.pdf", h = 5, w = 5, poin
ggplot() + geom_point(aes(y = value, x = age, color = sex),data = suball,size = 0.5) + geom_
  values = c("#F8766D","#00BFC4"),
```

```

  labels = c( "Female","Male")
)
dev.off()

```

pdf
2

```

#
# type <- "BD"
# sub <- scott_long %>% filter(BLMD == type) %>% arrange(age)
# y <- scott_long %>% filter(BLMD == type) %>% arrange(age) %>% pull(value)
# sub$iso_y <- pava(y)
# ggplot(aes(x = age, y = value), data = sub) + geom_point() + geom_path(aes(x = age, y = iso_y))
#
# type <- "LM"
# sub <- scott_long %>% filter(BLMD == type) %>% arrange(age)
# y <- scott_long %>% filter(BLMD == type) %>% arrange(age) %>% pull(value)
# sub$iso_y <- pava(y)
# ggplot(aes(x = age, y = value), data = sub) + geom_point() + geom_path(aes(x = age, y = iso_y))
#
# type <- "LD"
# sub <- scott_long %>% filter(BLMD == type) %>% arrange(age)
# y <- scott_long %>% filter(BLMD == type) %>% arrange(age) %>% pull(value)
# sub$iso_y <- pava(y)
# ggplot(aes(x = age, y = value), data = sub) + geom_point() + geom_path(aes(x = age, y = iso_y))

```

#Comparing scott and DER

```

#sum the scott scorres
#Average the dentin across raters.
#merge the scottt and DER data
#Plot scott vs DER.

scott_long_sum <- scott_long %>% group_by(id, castost, age, sex) %>% summarize(value = sum(value))

```

`summarise()` has grouped output by 'id', 'castost', 'age'. You can override using the `.groups` argument.

```

dentin_long_sum <- dentin_long %>% group_by(id, age) %>% summarize(DER = mean(DER))

```

``summarise()`` has grouped output by 'id'. You can override using the ``.groups`` argument.

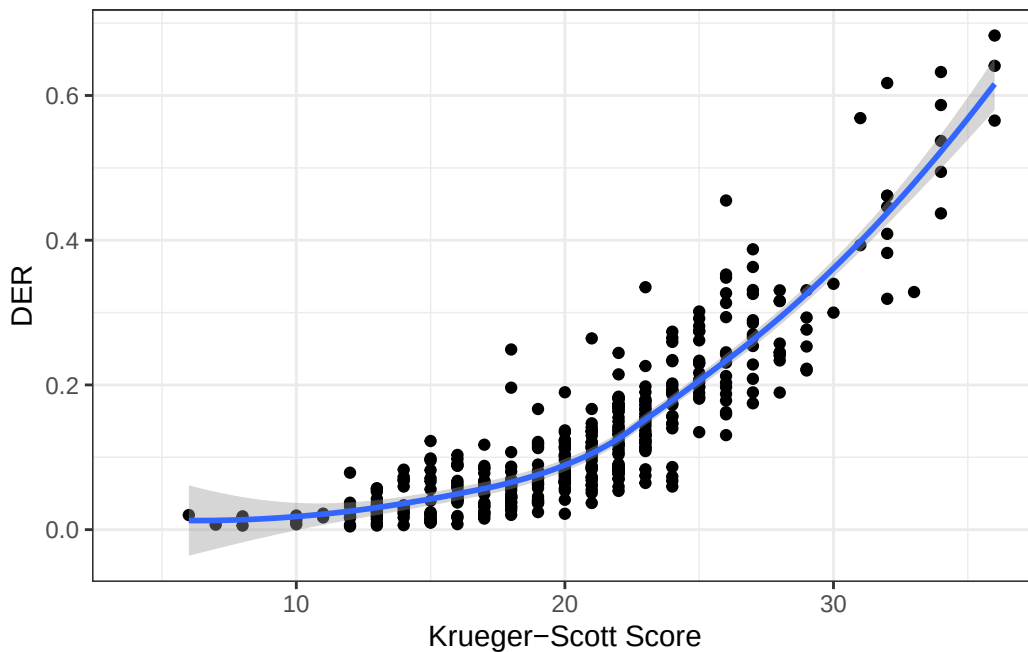
```
merg <- scott_long_sum %>% left_join(dentin_long_sum, by = c("id", "age"))
```

```
ggplot(aes(x = value, y = DER), data = merg) + geom_point() + xlab("Krueger-Scott Score") + g
```

``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

Warning: Removed 201 rows containing non-finite outside the scale range (``stat_smooth()``).

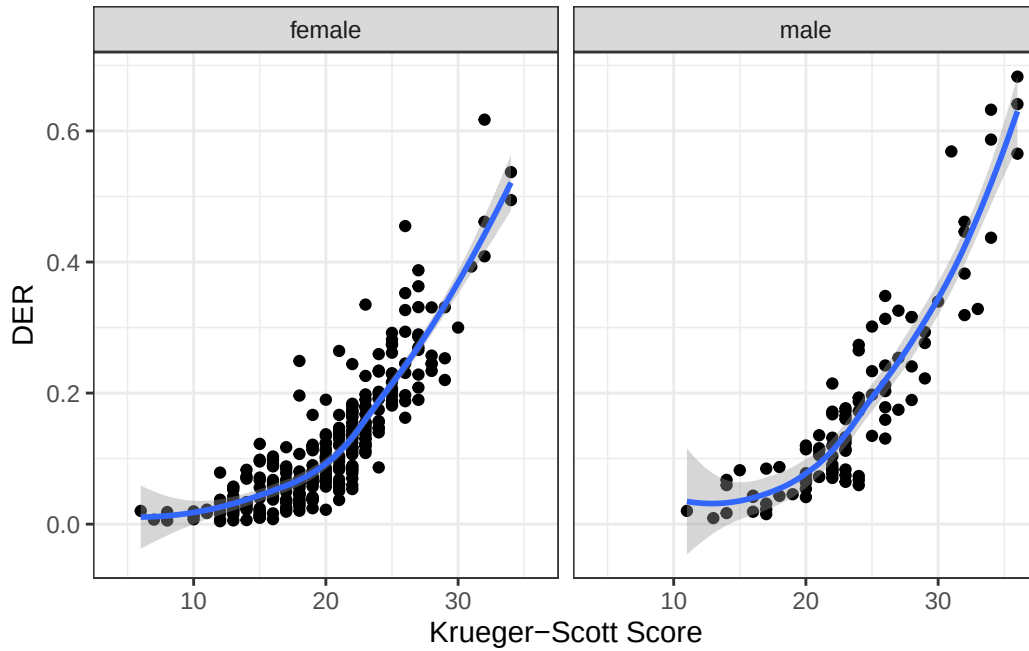
Warning: Removed 201 rows containing missing values or values outside the scale range (``geom_point()``).



```
ggplot(aes(x = value, y = DER), data = merg) + geom_point() + xlab("Krueger-Scott Score") + g
```

``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

Warning: Removed 201 rows containing non-finite outside the scale range
(`stat\_smooth()`).
Removed 201 rows containing missing values or values outside the scale range
(`geom\_point()`).



Notes for Greg

- <https://link.springer.com/article/10.1007/s11222-017-9797-8>
- <https://academic.oup.com/jrsssb/article/83/5/963/7056107>